Testing SWINGERsim

Part III: Eclipsing binaries with a changing inclination – V907 Sco, QX Cas and AH Cep

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Abstract. When an eclipsing binary has a third star on an inclined orbit, the orbit of the binary slowly turns (precesses) like a spinning top. Its inclination to our line of sight changes, so over decades the eclipses deepen, become shallower, or switch off and on. We test whether SWINGERsim, an N-body simulator for bound systems of three to ten stars, can take such a system from the published data, find a model that reproduces a century of observations, and say what comes next. Three binaries stand for three situations.

V907 Sco has a complete published solution (Zasche et al. 2023). From 58 TESS times of minima, 172 radial velocities and 14 inclinations measured since 1900, SWINGERsim finds a third star of 0.98 (+0.05/−0.04) $M_\odot$ on a 142.09 ± 0.04-d orbit, inclined by 25.8 ± 0.8° to the binary, with a nodal period of 65.5 (+0.9/−1.0) yr. All of these agree with the published values within about one standard deviation (the published ranges of m_3, the mutual inclination and the nodal period are 90 % limits). The model switches the eclipses off and on in the same years as the plates, the old photometry and TESS show. It predicts the depths of the 2021 and 2023 TESS eclipses to within 5 %, and it puts the end of the present eclipses in 2052–2057 (published: about 2049). The radial velocities of Lacy et al. (1999) fit the same 142-d orbit; a fit started at their 99-d period leaves it and ends in a clearly worse solution.

QX Cas stopped eclipsing in the 1990s, and its third star has never been seen (Guinan et al. 2012a,b). SWINGERsim shows which third stars can turn the orbit fast enough (most likely 0.5–3 years and 0.5–4 $M_\odot$) and that the inclination must have fallen at least twice as fast as the published linear rate. It also finds the binary in the TESS light curves of 2019–2024 through its light variation at the orbital period, at the amplitude the model predicts. Computing the eclipse **depth**, not only whether an eclipse is geometrically possible, shows a weak point: in our best model the eclipses of the 1930s are far too shallow for the photographic plates that recorded them. A model with deep eclipses in the 1930s exists, but it keeps the eclipses until the late 1990s, when none were seen. The old eclipse depths also require 13–47 % of third light, more than any third star of the model provides. Both point to the adopted stellar parameters, not to the dynamics.

AH Cep is a massive binary for which a change of the eclipse depth was once claimed. SWINGERsim reproduces 125 years of eclipse times with a third body on a 70-yr orbit of at least 6.6 $M_\odot$. Such a distant companion changes the inclination by only 0.02° in 400 years, so it cannot cause the claimed change. A fourth body that tilted the pair noticeably would show in the eclipse times or in the TESS eclipse depths, and the 9-yr fourth body once proposed would be dynamically unstable. The TESS light curves are best described by an inclination of about 72.50 ± 0.21° (published 69.8°) with about 5 % of extra light; this is in tension with the < 2 % third light allowed by the spectra and depends on the light-curve model.

Gaia DR3 shows no RUWE above 1.4 for the three systems. For V907 Sco, where our model moves the light centre by about 0.3 mas, this is a tension worth following up with Gaia DR4.

1 Aim

Earlier parts of the SQUADRA test checked SWINGERsim on compact multiple stars with complete published solutions. Here the question is harder: **given what has been measured over a century — a few inclinations, a list of eclipse times of very different quality,**

some radial velocities — can SWINGERsim find the system that produced it, check the published conclusions, and say something new?

For each system we ask four things:

1. What do the published papers conclude, and from which data?
2. What does SWINGERsim obtain from the same data?
3. Does SWINGERsim **confirm** the papers, **add** to them, or **disagree** with them, and why?
4. What does SWINGERsim predict that can be checked with data that already exist (here mostly TESS and Gaia)?

2 Data and methods

2.1 How SWINGERsim models a system

SWINGERsim describes a system as a hierarchy of orbits: here a close pair of stars (the eclipsing binary) and a third star orbiting the pair. From the masses, the orbits and the stellar radii and temperatures it integrates the motion of all stars with full gravity (N-body; REBOUND with the IAS15 integrator, Rein & Liu 2012, Rein & Spiegel 2015). The motion includes general relativity and the tidal and rotational deformation of the stars (REBOUNDx, Tamayo et al. 2020), which slowly turn eccentric orbits (apsidal motion). From the integrated motion SWINGERsim computes the times of the eclipses, the radial velocities of the stars, the inclination of the pair at any date, and synthetic light curves in a given passband (here TESS, Johnson B and V). The eclipse depths follow from the geometry and the stellar surfaces (spheres, or Roche surfaces for close stars).

2.2 Terms used in this report

Several words are used in a precise sense. They are defined here once.

- **Time of minimum (eclipse time).** The moment of the deepest light during an eclipse, measured by an observer from a light curve. These are **data**. They come from the literature, from the O − C gateway of the Czech Astronomical Society (var.astro.cz), from the O − C files of R. H. Nelson (AAVSO), and, for TESS, from our own measurement: for every TESS sector a mean eclipse profile is fitted to each eclipse with a free time.
- **Model eclipse time.** For every observed minimum the model computes the moment when the two stars of the pair are aligned with the observer (the conjunction), found in the N-body integration, plus the time the light needs to reach the observer from the pair's position on its orbit around the third star. It is computed at the epoch of every observed minimum and compared with it; it is **not a prediction** but the model's version of the same event. The geometric conjunction and the photometric minimum differ by seconds for these systems (about 1 s for AH Cep, from synthetic light curves), far below the errors of the data.
- **O − C** ("observed minus calculated"). The difference between an eclipse time and a strictly periodic clock (a linear ephemeris). It shows by how many minutes the eclipses come early or late. The model's O − C uses the same clock, so data and model can be drawn in one diagram.
- **Prediction.** A model eclipse time, inclination or light curve at a date **without data**. In this report predictions are only made outside the observed intervals, and they are marked as such.
- **Eclipse limit.** A pair with radii R_1, R_2 eclipses only while the projected distance of the two stars at conjunction is smaller than $R_1 + R_2$, i.e. while $\cos i < (R_1 + R_2)/r$, where r is their separation at conjunction. For V907 Sco this is $i > 72.8°$, for QX Cas $i > 70.7°$ (it changes slowly because the eccentric orbit of the pair turns), for AH Cep $i > 50.4°$. Above the limit the stars **can** eclipse; this does **not** mean that the eclipse was deep enough to be observed. Near the limit the eclipses are grazing and very shallow. Throughout the report we therefore separate a **geometric eclipse** (the inclination is above the limit) from a **detectable eclipse** (its depth is larger than what the photometry of the time could see: roughly 0.1 mag for photographic plates, a few hundredths of a magnitude for photoelectric photometry, a few tenths of a percent for TESS).

- **Third light.** Any light that does not take part in the eclipse: a third star of the system, a star in the same photometric aperture, or scattered light. If a fraction l_3 of the total light is third light, an eclipse that would remove a fraction d of the binary's light removes only $(1 - l_3)$ d of the total, and is shallower in magnitudes. The depth of an eclipse therefore always depends on three things together: the inclination, the sizes and surface brightnesses of the two stars, and the third light. Whenever this report gives an eclipse depth, it states which third light was assumed.
- **Nodal period.** The time in which the orbit of the pair turns once around the total angular momentum of the system. During one turn the inclination of the pair swings between two extremes.

2.3 Inclinations above 90°

A light curve measures only sin i: an orbit inclined by i and one inclined by 180° − i give the same eclipses. Published inclinations are therefore always ≤ 90°, and the main panel of every inclination figure shows the model in this **observable** form (0–90°). When the orbit passes through the edge-on position (90°) during its precession, the observable inclination rises to 90° and falls back, and then rises again once the orbit is past its turning point and turns back through edge-on. Over one passage the observable curve therefore has two peaks at 90° with a dip between them, an "M" shape.

The N-body model also knows on which side of the sky the orbit is. The second panel shows the same curve **unfolded** to 0–180°: there it is a single smooth wave, and the dip of the "M" turns out to be the true maximum of the inclination above 90°. Both panels contain the same information. The observable one is what a light curve can be compared with; the unfolded one shows the precession.

2.4 How SWINGERsim fits data

The fit varies the masses and orbits of the N-body model until the model matches all data at once:

- times of minima. Every source (visual, photographic, photoelectric, CCD, each TESS sector) gets its own extra error, found by the fit, and outliers beyond 4–5σ are removed;
- radial velocities (each data set with its own zero point and extra error);
- inclinations from published light-curve solutions at their epochs. One-sided information is used as well: "the inclination was below 75°" or "there were no eclipses in this year" (the model must then stay below its own eclipse limit);
- light curves, where stated.

Without astrometry only the difference of the nodes of the two orbits on the sky is measurable, so the node of the pair is fixed.

Errors. Unless stated otherwise the errors are from the curvature of χ^2 around the best fit (linear errors). For V907 Sco the posterior was also sampled with emcee (Foreman-Mackey et al. 2013; 48 walkers × 1000 steps, the first 500 steps discarded as burn-in). The autocorrelation time of the chain is $\tau \approx 54$ steps, so the whole chain covers about 19 τ and the 500 kept steps about 9 τ, i.e. about nine independent samples per walker and about 430 in total. Foreman-Mackey et al. (2013) suggest running for about ten τ, which the kept part nearly reaches. That is enough for the 16–84 % ranges quoted here, but not for the 50 τ that the emcee online documentation (tutorial "Autocorrelation analysis & convergence") recommends for a reliable estimate of τ itself. A chain of 400 steps gave errors that differ from those of 1000 steps by a median of 10 % (at most a factor of 1.5, for m_3 and the phase of the outer orbit), and medians that differ by less than 0.8σ. The quoted errors are therefore reliable to a few tens of percent. 100 samples of this posterior were integrated over 1890–2070 to give the ranges of the predicted inclination and of the eclipse switches.

Where the data are too poor for a unique fit (QX Cas), and for the question of a fourth body (AH Cep), SWINGERsim builds an **ensemble**: thousands of possible systems drawn from broad ranges are integrated, systems unstable by the criterion of Mardling & Aarseth (2001)

are discarded, and the rest are weighted by how well they reproduce the measurements. A single fit is one possible system; the ensemble shows the range of systems the data allow.

2.5 The systems and their data

Table 1. The three systems and the data used.

	V907 Sco	QX Cas	AH Cep
eclipsing pair	3.78 d, e ≈ 0.01, 2.74 + 2.56 $M_\odot$, late B (Zasche et al. 2023)	6.00 d, e ≈ 0.21, ≈ 5.5 + 6.5 $M_\odot$, B1.5 V + B3 IV (Guinan et al. 2012a)	1.775 d, circular, 16.1 + 13.7 $M_\odot$, B0.5 V (Pavlovski et al. 2018)
third body	142 d, e = 0.47, 1.06 $M_\odot$ (Zasche et al. 2023)	not detected	light-time orbit of ~68 yr (Kim et al. 2005)
inclinations	14 epochs 1900–2022 (Zasche et al. 2023, Table 2)	~1960, ~1980, no eclipses from 2003 (Guinan et al. 2012a)	~1983 (Pavlovski et al. 2018)
eclipses seen	1899–1918, 1963–1986 (Lacy et al. 1999), since 2019 (TESS)	photographic minima 1933–1974; photoelectric ~1960 and ~1980; none in the mid-1990s, 2003–2008 (Guinan et al. 2012a,b) and TESS 2019–2024	always
times of minima	58 from TESS 2021–2025	96 from 1933–1974 (74 photographic)	325 from 1901–2026 (216 TESS)
radial velocities	172: CHIRON 2022 (Zasche et al. 2023), 1983–1994 (Lacy et al. 1999)	–	–
light curves	TESS 2021, 2023	TESS 2019–2024 (no eclipses)	TESS 2019, 2024

TESS data (Ricker et al. 2015) are SPOC light curves unless stated otherwise.

2.6 How to read the pictures of the systems

Each system is shown in three panels (Figs. 1, 8, 12), the published system dashed and orange, the SWINGERsim result solid and blue:

- **From above.** The orbits seen from a point above the system, with Earth towards the bottom of the panel. The pair (label “pair m1+m2”) and the third star move around the common centre of mass.
- **The eclipsing pair on the sky.** What a telescope with unlimited resolution would see: the orbits of the two stars of the pair projected on the sky, with the stars drawn to scale. An eclipse happens when one disc passes in front of the other. The flatter the projected orbit, the closer we look along the orbital plane and the deeper the eclipses.
- **Where the pair’s orbit points.** The axis of the pair’s orbit (a unit vector) projected on the sky. A point at distance sin i from the centre means inclination i; the rim (distance 1) means that we look exactly along the orbital plane. Eclipses are possible only while the axis lies in the yellow band between the rim and the dotted circle of radius sin(eclipse limit). As the orbit precesses, the axis moves around the total angular momentum of the system (×); labels are years, and the large dot marks 2025. The panel is zoomed on the track.

3 V907 Sco

3.1 What the papers say

V907 Sco is a pair of late-B stars on a 3.78-d orbit with a solar-type third star. Its eclipses were seen on Harvard plates from 1899 to about 1918, again from 1963 to about 1986, and not in between (Lacy et al. 1999). Lacy et al. (1999) explained this by a third star on a 99.3-d orbit and a nodal period of about 68 years, and they predicted the return of the eclipses for 2030 ± 5. TESS saw shallow eclipses already in 2019 (sector 13) and deeper ones in 2021 (Borkovits 2022; Zasche et al. 2023). Zasche et al. (2023) measured radial velocities of all three stars in 2022 with CHIRON and found the outer orbit to be **142.01 ± 0.05 d, e_2 = 0.47 ± 0.02, m_3 = 1.06 (+0.11/−0.10) $M_\odot$** (the mass with 90 % confidence limits). They first fitted the 14 inclinations of the pair, 1900–2022, with a secular (orbit-averaged) model, but found the best fit "formally … statistically not acceptable", because the eccentric outer orbit adds long-period perturbations that the secular model lacks. They therefore used a fully numerical point-mass three-body model (their Sect. 3.2), constrained by the secular solution, and derived a mutual inclination of **26.2 (+2.6/−2.2)°**, a nodal period of **63.5 (+3.3/−2.6) yr** (both again 90 % confidence limits) and the end of the present eclipsing phase around **2049**. They did not use the older velocities of Lacy et al.

The question for SWINGERsim: can an N-body fit of all these data, including the old velocities, reproduce the switching eclipses, and does it confirm the third star and its orbit?

3.2 The system: published and fitted

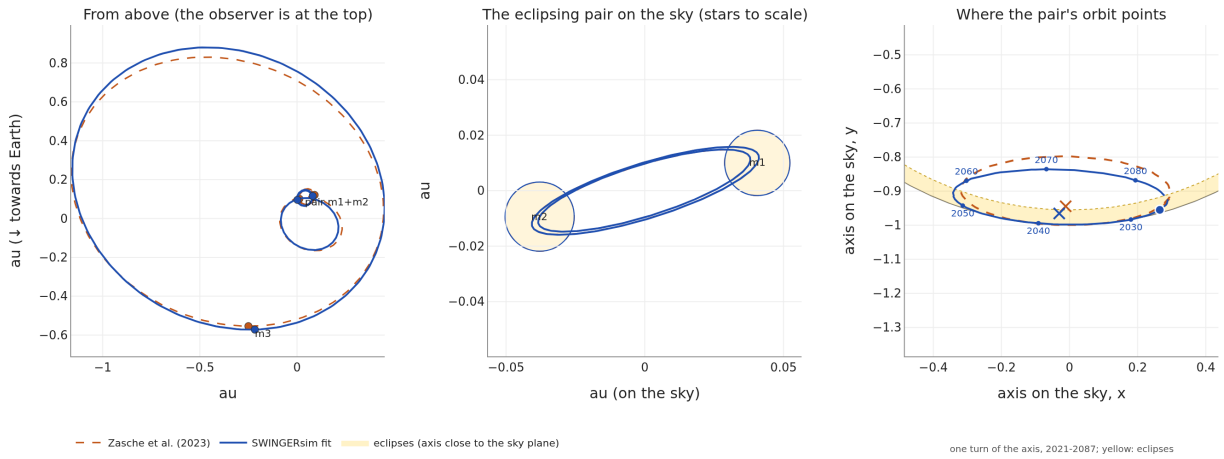


Figure 1. V907 Sco according to Zasche et al. (2023) (orange, dashed) and to the SWINGERsim fit (blue). Left: the orbits from above. Middle: the eclipsing pair on the sky in 2021, stars to scale. Right: where the axis of the pair's orbit points during one turn, 2021–2087; eclipses are possible while it is in the yellow band.

How to read Figure 1. The two systems are nearly identical. The third star (m3) moves on the large eccentric orbit; the pair moves on the smaller ellipse around the common centre of mass. In the middle panel the two discs just touch on the sky at their closest projected approach: in 2021 the pair eclipses only partially (inclination 76.6°, eclipse limit 72.8°). In the right panel the axis of the pair's orbit circles the total angular momentum (×) once in about 66 years. It enters the yellow band in 2019, stays in it until the mid-2050s, and then leaves it for about 30 years.

3.3 The inclination over 180 years

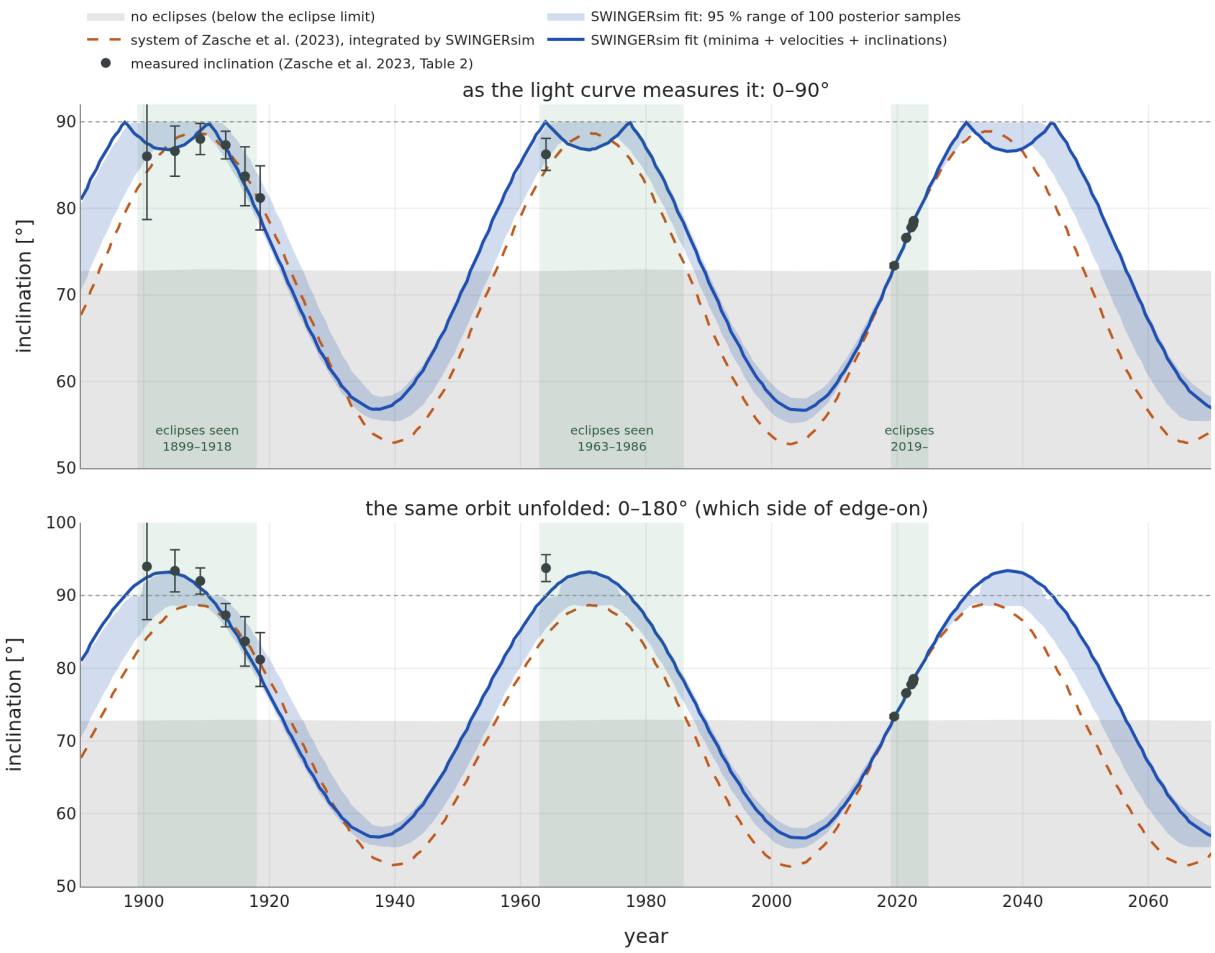


Figure 2. Inclination of the 3.78-day pair, 1890–2070. Top: as a light curve measures it (0–90°); bottom: the same model unfolded to 0–180°. Points: the 14 measured inclinations (Zasche et al. 2023, Table 2; in the bottom panel drawn on the side of 90° where the model is). Blue line and band: the SWINGERsim fit and the 95 % range of 100 posterior samples. Orange dashed: the published system integrated by SWINGERsim. Grey: below the eclipse limit. Green bands: when eclipses were observed (Lacy et al. 1999; TESS).

How to read Figure 2. Both panels show the inclination of the pair computed by the N-body model. Wherever the curve is above the grey area, the pair can eclipse; where it dips into the grey, the eclipses stop. **Top panel:** this is what a light curve measures and what the published inclinations are. In 1897–1911, 1964–1977 and 2031–2045 the curve has the "M" shape described above: the orbit passes edge-on (90°), turns back, and passes edge-on again. **Bottom panel:** unfolded, the same curve is one wave between about 57° and 93°. This is the precession of the orbit: the axis of the pair moves on a cone around the total angular momentum, so the inclination swings by twice the cone angle around the inclination of the total angular momentum (about 75°). Its period, the nodal period, sets how often the eclipses come and go. The fitted orbit passes through 90° in each cycle; the published one stays just below it. The 14 measured values cannot tell the two apart, because a light curve gives the same result for i and 180° − i.

The model curve passes through the measured points and through all observed changes: eclipses on the Harvard plates until 1918, eclipses in 1963–1986, the end seen by Lacy et al., and the rising inclination seen by TESS and FRAM since 2019. The band is narrow where data exist and widens towards the past and the future; the fit and the published model separate after 2040. The published errors of the TESS and FRAM inclinations (0.02–0.10°, and 0.25°

for the shallow eclipses of TESS sector 13) are mostly smaller than the differences between light-curve models (see the eclipse depths below), so the fit uses an error of at least 0.1°. With it the 14 inclinations give $\chi^2 = 13.7$ for the fit and 6.0 for the published system.

Table 2. V907 Sco: when the eclipses switch. "Geometric" is the moment when the model inclination crosses the eclipse limit; the model eclipse depth in V (third light of the third star included, 1.7 %) shows when the eclipses became too shallow to be detected.

change	observed	published system in SWINGERsim	SWINGERsim fit, geometric (median, 95 % range)	SWINGERsim fit: model depth in V
eclipses stop	~1918 (Lacy et al. 1999; last DASCH inclination 1918)	1923.5	**1923.5** (1921.7–1925.5)	0.16 mag in 1918, 0.06 mag in 1920
eclipses return	seen from ~1963 (Lacy et al. 1999; reported by Strohmeier et al. 1964)	1956.5	**1954.0** (1952.0–1955.5)	0.08 mag in 1955, 0.28 mag in 1958
eclipses stop	~1986 (Lacy et al. 1999)	1986.5	**1989.0** (1988.0–1990.0)	0.11 mag in 1986, 0.02 mag in 1988
eclipses return	2019, depth 0.6 % (TESS S13)	2019.5	**2019.5**	0.002 mag in mid-2019 (eclipse alone; with the light variation outside eclipses 0.008 mag)
eclipses end	~2049 (Zasche et al. 2023)	2050.0	**2054.5** (2052.5–2056.8)	0.19 mag in 2052, 0.03 mag in 2055

Photographic and photoelectric photometry of the time could see eclipses deeper than about 0.05 mag. The geometric crossing comes 1–3 years after the model eclipses become shallower than this, so the model and the observations agree on all four observed changes. No eclipses were recorded between 1925 and 1960, in agreement with the model; the model switches them on in the mid-1950s, a few years before the first reported observation (1963; Strohmeier et al. 1964 reported them as new). When the published parameters are integrated by SWINGERsim, the switches and the end of the eclipses (2050.0; 0.014 mag in 2049) agree with the published numerical point-mass model (about 2049). This is an independent check of SWINGERsim's N-body integration.

3.4 The eclipse times

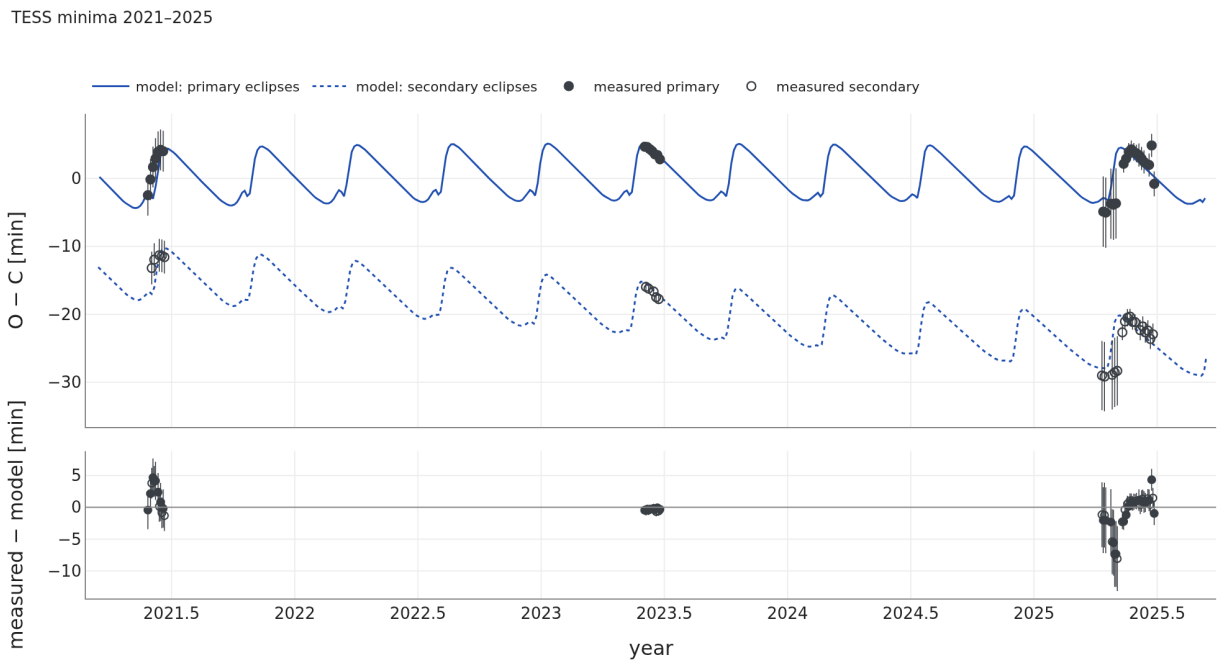


Figure 3. TESS times of minima 2021–2025 against a fixed period (top) and minus the model (bottom). Lines: model eclipse times; points: measured minima, filled for primary and open for secondary eclipses.

How to read Figure 3. The top panel is an O − C diagram: it shows by how many minutes the eclipses come early or late compared with a strictly periodic clock. The lines are the model eclipse times (conjunctions in the N-body model plus light-travel time), computed for every observed eclipse and continued between them. The saw-tooth repeats with the 142-d outer period. The primary and secondary eclipses drift apart (from 13 to 28 min in four years) because the third star slowly changes the eccentricity and the orientation of the pair's slightly eccentric orbit ($e \approx 0.011$): the secondary eclipse moves relative to the primary one. The measured minima lie on the model; the residuals (bottom) are 2.6 min rms, and the primary and secondary minima agree within 0.45 min (0.7σ). The larger scatter in 2025 comes from the lower-quality QLP photometry of sectors 91–93.

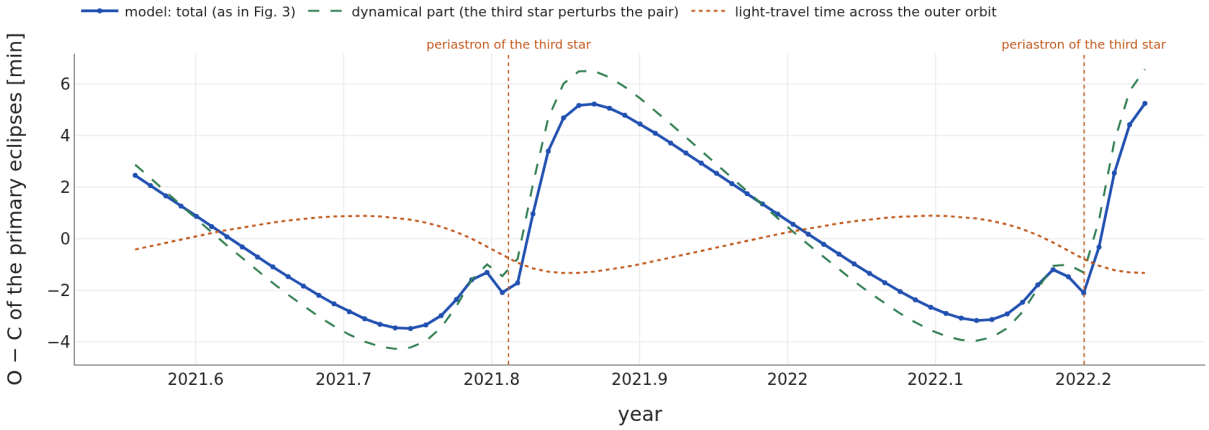


Figure 4. What the eclipse-time signal of V907 Sco is made of, one outer orbit in 2021–2022: the model O − C of the primary eclipses (blue), its dynamical part (green: the eclipses as they happen at the pair) and the light-travel time across the outer orbit (orange). Dotted vertical lines: periastron passages of the third star.

How to read Figure 4. The model O − C has two parts.

- **The light-travel time** (orange): the pair moves on its orbit around the common centre of mass with the third star, so its light needs more or less time to reach us. For V907 Sco this is small: ±1.1 min.
- **The dynamical part** (green, 10.8 min peak to peak) dominates by a factor of about five, as Zasche et al. (2023, Sect. 4 and their Fig. 11) found both with analytic formulae and with a direct numerical integration. The third star's tidal pull stretches the pair's orbit and lengthens its period, the more the closer the third star is. On the eccentric outer orbit ($e_2 = 0.47$) the third star comes to 0.53 au at periastron, only about six times the size of the pair's orbit, and the pair's period is then longest. The eclipses therefore fall behind quickly for two to three weeks after each periastron (the steep rise of the saw-tooth). For the rest of the outer orbit the period is shorter than the mean, and the eclipses gradually come earlier (the slow decline).
- **The small notch just before each rise** is a single eclipse that comes about half a minute early. During the few days of the periastron passage the pull of the third star changes within one revolution of the pair, so it shifts the eclipse that falls closest to periastron differently from its neighbours. This is a property of the N-body model, not a numerical artefact: with a five times finer sampling of the integration the notch changes by less than 0.02 min. It is too brief (one eclipse) and too small to be checked with the present minima.

3.5 The radial velocities and the 99-day question

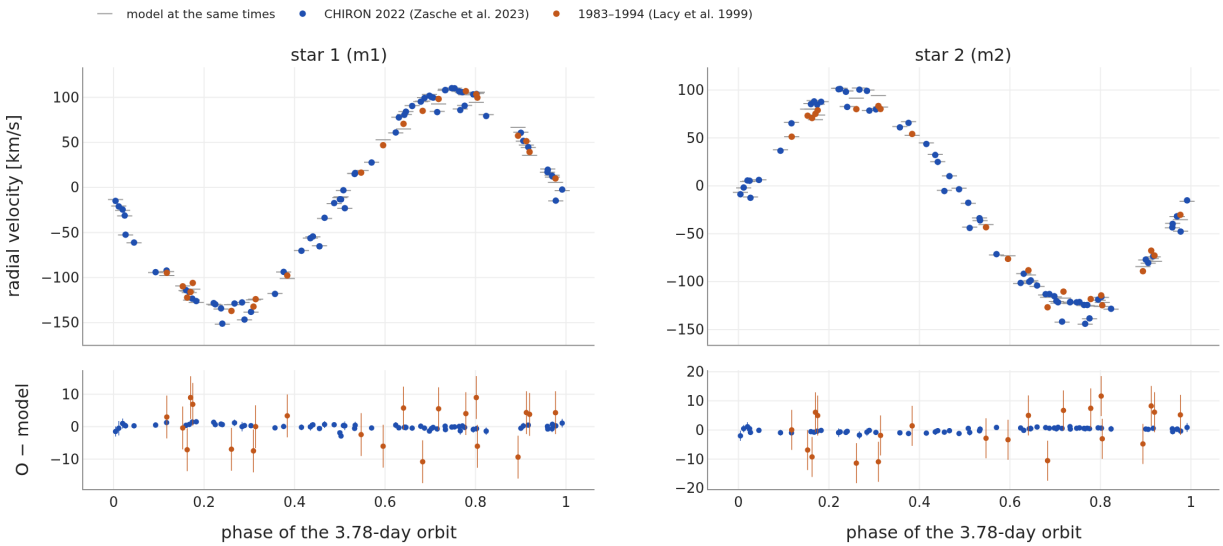


Figure 5. Radial velocities of the two stars of the pair against the phase of the 3.78-day orbit. Blue: CHIRON 2022 (Zasche et al. 2023); orange: 1983–1994 (Lacy et al. 1999); grey ticks: the model at the times of the observations; bottom: measured minus model.

How to read Figure 5. Each star moves towards us and away from us once per orbit, which gives the sine-like curves. The grey ticks are the model at the same moments. They include the pair's motion on the outer orbit, which shifts groups of points up or down by up to ±13 km/s; this is why the ticks do not lie on one curve. The phase is computed from the model's own eclipses near each observation, because the light-time effect and the precession shift the phase between 1983 and 2022. The model fits the CHIRON velocities to 0.8 km/s rms. The velocities measured 30–40 years earlier scatter by 6.6 km/s around the same model, which is consistent with their precision.

The velocities of Lacy et al. (1999) fit the 142-d orbit of Zasche et al. (2023) after a zero-point shift of -1.8 ± 1.1 km/s. The two data sets together tie the phase of the outer orbit over about 80 outer cycles. To test the 99-d period directly, a second fit was started at $P_2 = 99.3$ d, $e_2 = 0.3$, without priors and with P_2 free between 60 and 200 d. It left 99 d and stopped in a local minimum at $P_2 = 105.9$ d with $m_3 = 0.37\ M_\odot$. That solution fits the CHIRON velocities only to 7.1 km/s rms (the quoted errors are about 0.7 km/s) and the best TESS minima twice worse; its likelihood is lower by $\Delta(-2 \ln L) \approx 580$. **The data rule out the 99-d outer orbit and confirm the 142-d one.**

3.6 The eclipse depths: a prediction tested

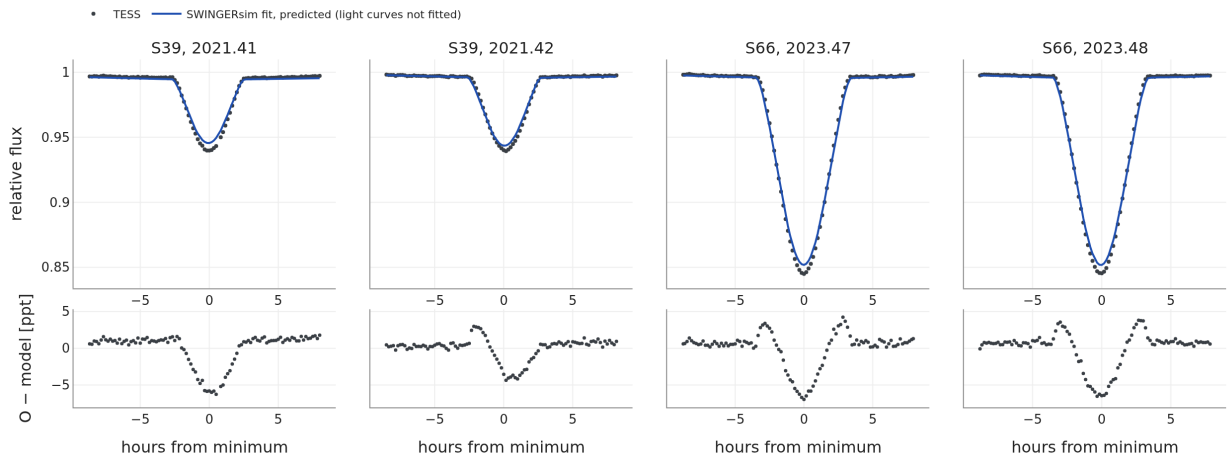


Figure 6. TESS eclipses of 2021 (sector 39) and 2023 (sector 66) with the light curve predicted by the SWINGERsim fit, which did not use the light curves (only a zero point and a slope are adjusted per eclipse); bottom: residuals in parts per thousand (ppt).

How to read Figure 6. The fit used eclipse times, velocities and the 14 published inclinations, but no light curves. Its synthetic light curve, with the radii and temperatures of Zasche et al. (2023) and the third light of the third star (1.7 % in TESS, from its mass), predicts the growth of the eclipses from 5.3 % (observed 5.5 %) in 2021 to 14.3 % (observed 15.0 %) in 2023. This is a real prediction: the depths were not fitted. The residuals have a "W" shape: the measured eclipses are about 5 % deeper and slightly shorter than predicted.

When the TESS light curves are added to the fit, the fit removes this difference by raising the present inclination by about 0.3° and shrinking the second star (R_2 = 2.51 instead of 2.68 $R_\odot$). The 14 published inclinations, which Zasche et al. derived with PHOEBE from the same kind of light curves, are then matched much worse (χ^2 = 222 instead of 13.7). SWINGERsim's synthetic eclipses are shallower than PHOEBE's for the same parameters, by about 5 % in depth (see also AH Cep); the published inclinations and SWINGERsim's depths therefore differ by a fraction of a degree for the same light curve. We use the fit without light curves as the result; the switch years and the end of the eclipses change by less than two years between the two fits.

3.7 Comparison

Table 3. V907 Sco: the published solution and SWINGERsim. Published ranges of m_3, the mutual inclination and the nodal period are 90 % confidence limits (Zasche et al. 2023, Sect. 3.4). SWINGERsim: posterior median and 16–84 % range of the MCMC (1000 steps), i.e. about 1σ.

quantity	Zasche et al. (2023)	SWINGERsim
mass of the third star m_3	1.06 (+0.11/−0.10) $M_\odot$ (90 % C.L.)	**0.98 (+0.05/−0.04) $M_\odot$**
outer period P_2	142.01 ± 0.05 d	**142.09 ± 0.04 d**
outer eccentricity e_2	0.47 ± 0.02	**0.467 (+0.010/−0.013)**
mutual inclination	26.2 (+2.6/−2.2)° (90 % C.L.)	**25.8 ± 0.8°**
nodal period	63.5 (+3.3/−2.6) yr (90 % C.L.)	**65.5 (+0.9/−1.0) yr** (the published system in SWINGERsim: 63.2 yr)

quantity	Zasche et al. (2023)	SWINGERsim
14 inclinations: χ^2	6.0 (published system integrated by SWINGERsim)	13.7
end of the present eclipses	~2049	**2052–2057** (95 %)
TESS minima: rms	–	2.6 min (58 minima)
radial velocities: rms	–	CHIRON 0.8 km/s, Lacy et al. 6.6 km/s
outer period from the 1983–1994 velocities	99.3 d (Lacy et al. 1999)	142.1 d; 99 d ruled out

How to read Table 3. Each row compares one quantity; the bold values are SWINGERsim's. The two columns do not quote errors at the same confidence level. For m_3, the mutual inclination and the nodal period, Zasche et al. give 90 % limits, about 1.6 times wider than 1σ limits for a Gaussian distribution, while SWINGERsim's ranges are 16–84 % (1σ). Scaled to the same level, the published ranges correspond to roughly ±0.06–0.07 $M_\odot$, ±1.3–1.6° and +2.0/−1.6 yr. With this correction the differences (−0.08 $M_\odot$, −0.4° and +2.0 yr) are still about one combined standard deviation or less. SWINGERsim's ranges are narrower; its fit also uses the TESS eclipse times and the velocities of Lacy et al. (1999), which the published long-term model did not.

3.8 Verdict

- **Confirms** the third star, its 142-d orbit, the mutual inclination and the precession of Zasche et al. (2023); every quantity agrees within about one standard deviation once the different confidence levels (90 % in the paper, 16–84 % here) are taken into account. From the dynamics alone the N-body model reproduces 120 years of inclinations, all four observed switches of the eclipses and the growth of the eclipse depth in 2021–2023.
- **Adds:** the radial velocities of Lacy et al. (1999) fit the same 142-d orbit, and the 99-d outer period they proposed is ruled out. The old and the new velocities together fix the phase of the outer orbit over 40 years.
- **Adds:** the drift of the primary and secondary eclipses in TESS 2021–2025 is the change of the pair's orbit caused by the third star, and the model reproduces it.
- **Differs slightly:** the nodal period of the fit is 2.0 yr longer than the published value but within its 90 % limits, and the present eclipses end later: in 2052–2057 instead of about 2049. The inclination keeps rising until about 2035 in both models; the eclipse depths of the next ten years will decide between them.
- **Can be checked with Gaia** (see "Independent checks" below): the third star moves the light centre of the system on the 142-d orbit by about 0.14 au, which Gaia should see.

4 QX Cas

4.1 What the papers say

QX Cas is a 6.0-d eccentric binary of two early-B stars in the young open cluster NGC 7790. Its eclipses, about 0.32 and 0.28 mag deep (Guinan et al. 2012b; the abstract does not name the band), were observed photoelectrically around 1960 (Sandage & Tammann 1969) and 1980 (Moffett & Barnes 1983). Older times of minima, 1933–1974, come from photographic plates and visual estimates. Guinan et al. (2012a, Figs. 2 and 3), adopting the PHOEBE light-curve solutions of Bonaro et al. (2009, published only as an abstract), show inclinations of about 85.4° around 1960 and 82° around 1980, and, from the FCAPT photometry of 2003–2008 (more than 110 nights, no eclipses), about 76° from 2003 onwards. They also note that photometry by A. Henden in the mid-1990s shows no eclipses. The change is attributed to precession driven by an unseen third body. No third star has been detected, and the papers

do not state whether third light was included in the light-curve solutions. A spectroscopic orbit (e = 0.22) by Kjurkchieva et al. (2007) is quoted in the AAVSO forum (Henden 2020), but we could not locate the publication.

The questions for SWINGERsim: what third star can turn the orbit fast enough; were the old eclipses deep enough to be seen, and with how much third light; and what should we see today, when the binary no longer eclipses?

4.2 Where the unseen third star can be

The ensemble asks which third stars can turn the orbit at the measured rate. 2000 systems were drawn with a third star of 0.3–6 $M_\odot$, an outer period of 30–11 000 d, an outer eccentricity of 0–0.6 and an arbitrary orientation; only systems stable by Mardling & Aarseth (2001) were kept. Each system was integrated 30 years forward and backward and weighted by how well it gives the inclination around 1980 (82 ± 1.5°) and its rate of change (−0.23 ± 0.08°/yr, the linear rate implied by Guinan et al. 2012a).

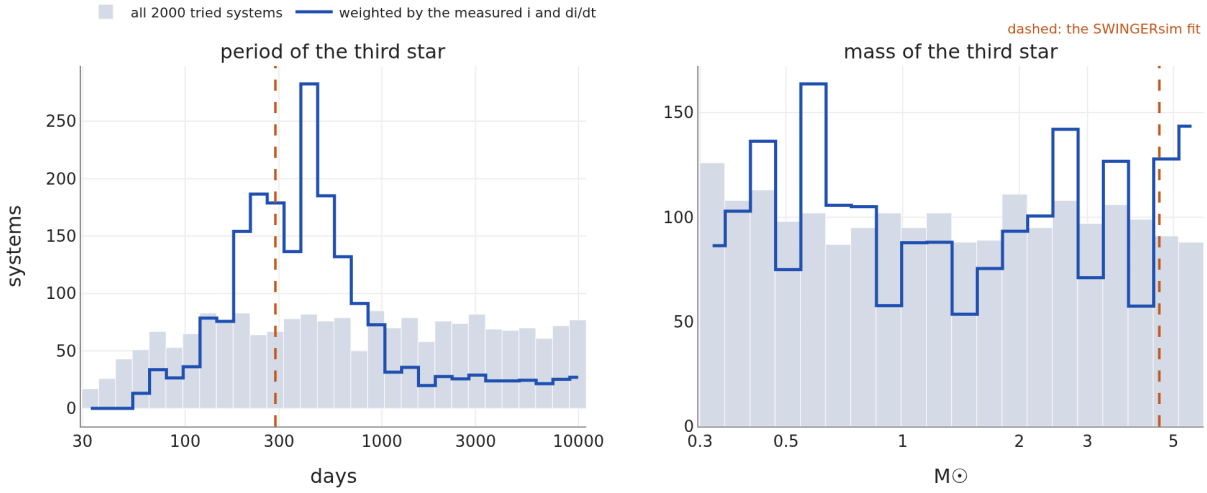


Figure 7. QX Cas: which third stars can turn the binary's orbit at the measured rate. Grey: the 2000 systems tried; blue: weighted by the measured inclination and its rate of change. Dashed: the SWINGERsim fit below.

How to read Figure 7. The grey histograms show what was tried (all periods and masses equally likely on a logarithmic scale). The blue line shows which of them can produce the observed precession. The precession rate grows with the mass of the third star and falls steeply with its distance (roughly as $m_3\ P_1/P_3^2$), so third stars closer than about 100 d turn the orbit too fast (or are unstable), and those farther than a few thousand days too slowly. The mass alone is hardly constrained, because a light star close in and a heavy star farther out produce the same rate.

Table 4. QX Cas: the third star according to the weighted ensemble (16 %, 50 %, 84 %; effective number of systems 187).

quantity	16 %	median	84 %
period P_3 [d]	193	414	1028
mass m_3 [$M_\odot$]	0.46	1.33	3.87
mutual inclination [°]	44	85	135

80 % of the weight lies at P_3 = 100–1000 d. In 13 % of all systems tried the two stars of the pair collided within 30 years; these had the third star's orbit nearly perpendicular, which

pumps the eccentricity of the pair (the von Zeipel–Lidov–Kozai effect). QX Cas has survived for tens of millions of years, so its third star is probably not on such an orbit.

4.3 A fit that knows the eclipses are gone

The fit used the 96 minima of 1933–1974 and eight pieces of inclination information: the two light-curve inclinations (~1960, ~1980), the upper limit of 75° for 2003–2008, "the pair eclipsed" in the 1930s, and "no eclipses" for 2003–2008 (FCAPT) and for TESS 2019, 2022 and 2024. Nine parameters were free (the pair's period, phase, eccentricity vector and inclination; the outer period, inclination and node; m_3), with broad priors $m_3 = 3 \pm 2\ M_\odot$ and $P_3 = 400 \pm 300$ d; the outer eccentricity was fixed at 0.2. "The pair eclipsed" was entered in its weakest form: the model inclination had to be above the eclipse limit, i.e. a geometric eclipse. The next section shows that this is not enough.

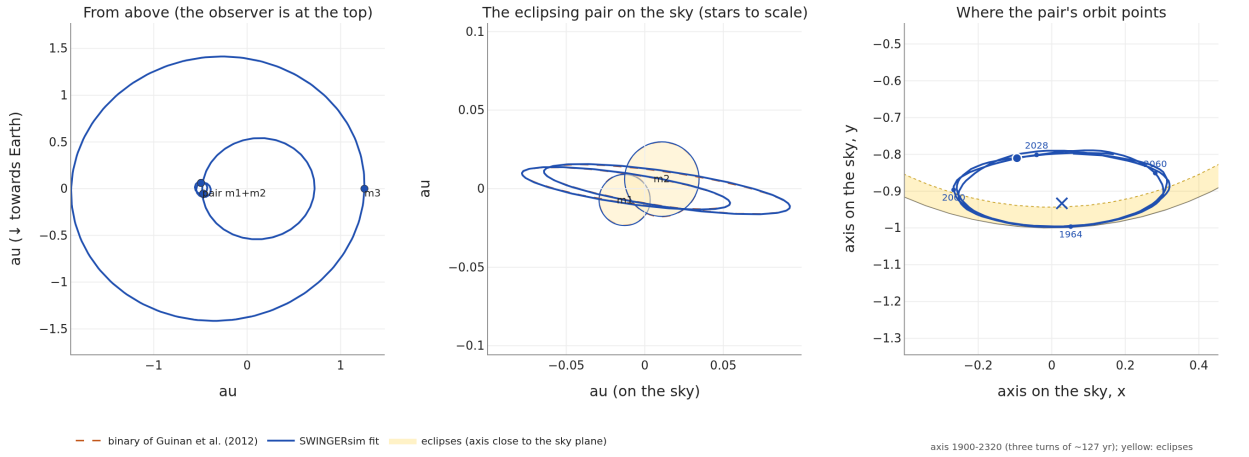


Figure 8. QX Cas: the binary as described by Guinan et al. (2012a) (orange, dashed; no third star) and the SWINGERsim fit with its third star (blue), in 1980. Right: the pair's axis 1900–2320 (three turns); eclipses are possible while it is in the yellow band.

How to read Figure 8. The published binary (orange) is drawn at the place of the fitted pair. The fit adds a third star of 4.6 $M_\odot$ on a 288-d orbit (large ellipse, left). The middle panel shows the pair in 1980, the reference date of the fit, at 81.8°: the discs overlap, so it eclipsed then. In the right panel the pair's axis circles the total angular momentum once in about 137 years; it left the yellow band around 2000 and returns to it around 2070. The track does not close exactly because the apsidal motion of the eccentric pair slowly changes the eclipse limit (the dotted circle).

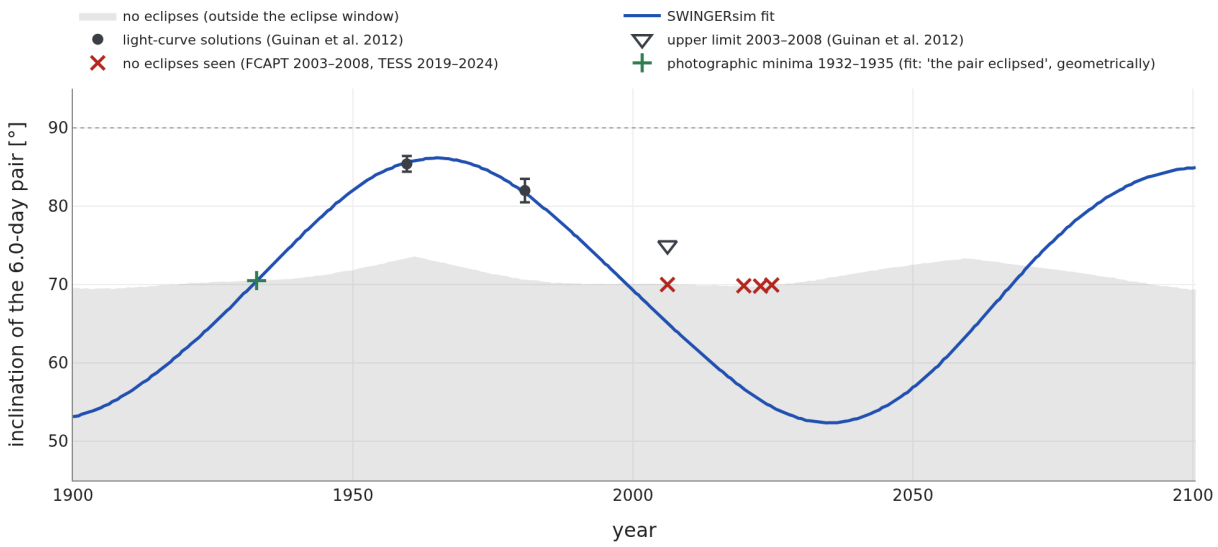


Figure 9. QX Cas: inclination of the 6.0-day pair, 1900–2100, in the SWINGERsim fit. Points: light-curve solutions (Guinan et al. 2012a); triangle: the upper limit 2003–2008; ×: no eclipses (FCAPT, TESS); +: the photographic minima of the 1930s, entered as "the pair eclipsed geometrically". Grey: below the eclipse limit.

How to read Figure 9. The fitted inclination stays below 90°, so the observable and the unfolded curves are the same. It rises through the eclipse limit in 1933, reaches 86° around 1965, falls through 81.8° in 1980 and leaves the eclipsing range in 1999. It satisfies all eight pieces of information ($\chi^2 = 0.1$). The upper edge of the grey area, the eclipse limit, is not constant: it depends on the separation of the stars at conjunction, which changes as the eccentric orbit of the pair turns (apsidal motion). The geometric eclipses return in 2071 in this fit; an MCMC (below) places the return anywhere between the early 2070s and after 2150.

Note the green cross. It sits exactly on the eclipse limit (model 70.58°, limit 70.59° in 1933): the fit has pushed the inclination of the 1930s just high enough for a geometric eclipse and no higher. That is a warning sign, and it is the subject of the next section.

4.4 Were the eclipses deep enough to be seen? Depth and third light

A geometric eclipse near the limit is grazing: one star covers only a thin edge of the other. To know whether an eclipse could have been **observed**, we computed its depth. For the stars of the fit ($R_1 = 3.5\ R_\odot$, 24 000 K; $R_2 = 5.1\ R_\odot$, 18 000 K, from the spectral types and the fractional radii of Guinan et al. 2012a) SWINGERsim calculated the light curve of the pair alone, with spherical stars and without the light variation outside eclipses, for inclinations 66–90° and for every orientation ω of the eccentric orbit. The depth then follows along the fitted inclination history. The light of a third star or of any other source was added as a fraction l_3 of the total.

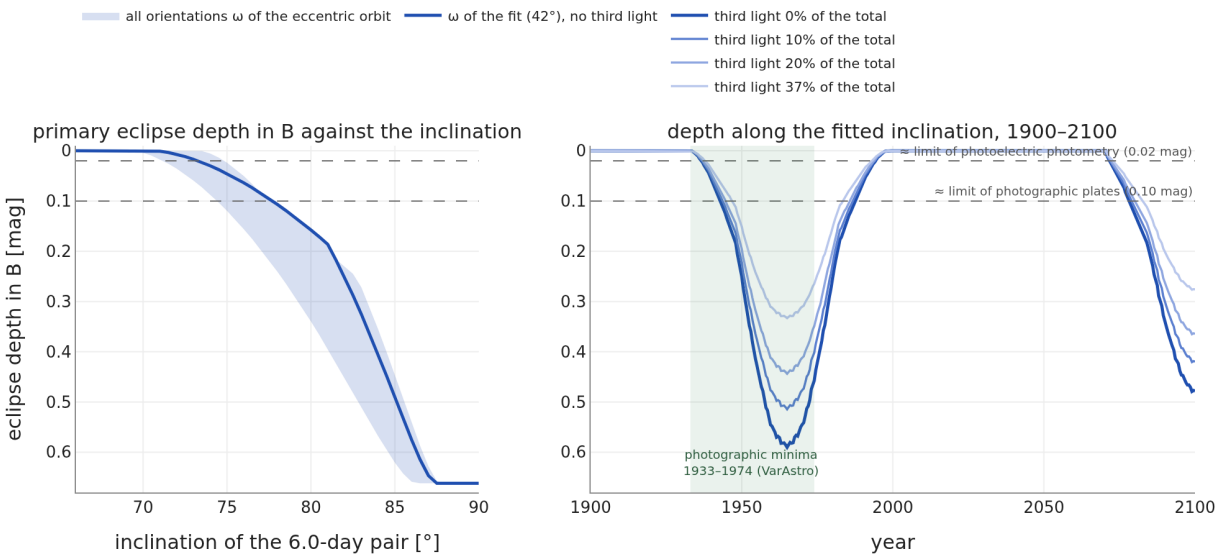


Figure 10. QX Cas: was an eclipse deep enough to be seen? Left: depth of the primary eclipse in B (close to the photographic band) against the inclination, for the stars of the fit without third light; band: all orientations ω of the eccentric orbit. Right: the depth along the fitted inclination history for third light of 0, 10, 20 and 37 % (ω of 1980). Dashed: the approximate detection limits of photographic plates (0.10 mag) and of photoelectric photometry (0.02 mag). Green: the years of the photographic minima.

How to read Figure 10. Left: a geometric eclipse begins at about 71°, but it stays shallower than 0.02 mag down to about 73° and reaches 0.1 mag only at **78°** (74.5–78° depending on ω; 79° with 20 % third light). Below about 87° the stars never cover each other fully, which is why the depth keeps growing up to that inclination. Right: along the fitted history the eclipses become detectable on photographic plates only from **1943** (1945 with 20 % third light) to 1987.5, and photoelectrically from 1937 to 1994.5. In 1933 the model eclipse is only **0.0006 mag** deep.

Table 5. QX Cas: the inclination needed for a detectable primary eclipse in B, for the stars of the fit (orientation ω of 1980; in brackets the range over all ω).

detection limit	no third light	20 % third light
photographic, ≈ 0.10 mag	78.0° (74.5–78.0°)	79.0° (75.5–79.0°)
photoelectric, ≈ 0.02 mag	73.5° (71.5–75.0°)	74.0° (71.5–75.5°)

This contradicts the data. VarAstro lists 11 photographic minima from 1933 to 1942 (Kippenhahn, Perova, Erleksova), the first in 1933. In the fitted model these eclipses would have been invisible on plates. So either the inclination of the 1930s was at least ~76–78°, higher than our fit gives, or these early minima are not what they seem (for example the times of minima of another star, or of a few plates with large errors). The fit has 288 days for the third star and a nodal period of 137 years; a higher inclination in the 1930s needs an earlier rise, i.e. a different phase of the precession or a faster one. The plates of the Harvard collection (DASCH), which cover NGC 7790 from about 1900 to 1950, could decide independently: an eclipse of 0.1 mag or more should be visible on them before 1943 if the minima of the 1930s are real. The next section asks whether a model can have deep eclipses in the 1930s at all.

The depths of the 1960s and 1980s need third light. At the published inclinations the model gives deeper eclipses than observed (V band, ω of 1980). Guinan et al. (2012b) do not say in which band the depths of the 1960s were measured; comparing them with V-band model depths is our assumption, and Table 6 is uncertain to that extent:

Table 6. QX Cas: model eclipse depths at the published inclinations, and the third light needed to make them as shallow as observed.

epoch, inclination	eclipse	model depth (no third light)	observed	third light needed
~1960, 85.4°	primary	0.50 mag	0.32 mag	31 %
~1960, 85.4°	secondary	0.33 mag	0.28 mag	13 %
~1980, 82°	primary	0.24 mag	0.12 mag	47 %

One value of third light cannot make both eclipses of 1960 as shallow as observed, and the value needed in 1980 differs from that of 1960. Third light alone therefore does not explain the depths; the ratio of the surface brightnesses of the two stars (their temperatures and radii) and the inclinations themselves must differ from what we adopted, as J. Kolář pointed out to us. The fitted third star of 4.6 $M_\odot$ would contribute about 12–14 % of the V light (MIST models, Choi et al. 2016, at the age of NGC 7790), which is of the right order but not a full explanation. The eclipse limit depends only on $R_1 + R_2$, so the geometric switching dates do not depend on the third light; the detectable dates do, by 1–4 years (Fig. 10, right).

4.5 A second fit: deep eclipses in the 1930s

To test whether the old photographic minima can be reconciled with the rest, we repeated the fit with the same 96 minima and priors, but entered the photographic eclipses as **detectable**: an inclination of at least 76 ± 1.5° in 1933 and in 1974, the value at which the eclipse of the fitted stars reaches about 0.1 mag in B (Table 5, with some allowance for ω and third light). All other inclination information is the same as before.

Table 7. QX Cas: the fit with detectable photographic eclipses (“deep 1930s”) compared with the fit above (“geometric 1930s”). Each row is one quantity; values are best fits with their linearised 1σ errors. Depths are in B for the stars of the fit without third light. *Detectable* means deeper than 0.1 mag (photographic) or 0.02 mag (photoelectric).

quantity	geometric 1930s (above)	deep 1930s
χ^2 of the nine inclination items	0.1	0.2
inclination in 1933	70.6° (at the eclipse limit)	75.8° (0.1σ below the required 76°)
inclination in ~1960, 1980	85.6°, 81.8°	85.7°, 81.5°
eclipse depth in 1933	0.0006 mag	0.069 mag
geometric eclipses	1933–1999, return 2071	1922–2005.5, return 2058
detectable on plates	1943–1987.5	1936.5–1989.5
detectable photoelectrically	1937–1994.5	until 1999.0
third star m_3, P_3	4.6 ± 0.6 $M_\odot$, 288 ± 1 d	4.56 ± 0.68 $M_\odot$, 287.8 ± 0.9 d
inclination of the outer orbit	–	72.5 ± 2.5°
rms of the 96 minima	98 min (photographic)	101 min (all sources)

The new fit reaches 75.8° in 1933, within 0.1σ of the requirement, and still satisfies the inclinations of 1960 and 1980, the upper limit of 2003–2008 and the absence of eclipses in 2003–2024 (model 69.4° in 2006 against the eclipse limit of 70.0°). The third star does not change. The eclipses of 1933 now have 0.069 mag in B: still shallower than 0.1 mag in that year, but detectable on plates from 1936.5, so most of the photographic minima of 1933–1942 fall into the detectable interval.

The price is the 1990s. To be high early, the inclination must rise earlier, and the whole history shifts: the geometric eclipses now last until 2005.5, and the eclipses stay deeper than 0.02 mag until 1999.0. A. Henden saw no eclipses in the mid-1990s (Guinan et al. 2012a). With the adopted radii and temperatures, deep eclipses in the 1930s and no eclipses in the mid-1990s pull against each other. They can be reconciled only if the eclipse depth at a given inclination is different from what our stars give, i.e. if the radii and the light ratio of the two stars (and the third light) are fitted together with the old depths. This is the same conclusion that the third light in Table 6 points to: the adopted stellar parameters are the weak link, not the dynamics.

What the ensemble says about the dates. Both fits are single systems. To obtain a range of dates we integrated the ensemble of Figure 7 (the 1654 of the 2000 systems that carry any weight; the other 346 have zero weight) over 1900–2100 and recorded when each system starts and stops eclipsing geometrically.

- **Weighted only by the inclination and its rate around 1980** (187 effective systems): 91 % of the weight eclipses in 1933. Only 15 % of the weight ends its eclipses between 1900 and 2025, in 1997–2011.5–2021.5 (16–50–84 %). Of those, 12 % return by 2100, in 2046–2058–2086. Most systems with the right inclination and rate in 1980 simply keep eclipsing: two numbers from 1980 do not determine the history.
- **Weighted also by the whole inclination history** (1960, 1980, the limit of 2005, no eclipses 2003–2024, and either form of the 1930s): only one or two effective systems remain. They end their eclipses in about 2002 and return around 2079.

The ensemble therefore contains almost no systems that satisfy the whole history: drawing the third star at random and weighting afterwards is too inefficient here, because the history is a narrow target in a nine-dimensional space. A reliable range of dates needs a search conditioned on the history, i.e. an MCMC of the fit itself with the inclination history as data (*Physics and Methods*, chapter Fitting observed data).

An MCMC of the deep-1930s fit. We sampled the posterior of the second fit with all its data: the 96 minima and the nine inclination items (emcee, 24 walkers × 6000 steps, the first 3000 discarded). The autocorrelation time is 311 steps, so the kept chain is 9.6 autocorrelation times long, just short of the ten usually asked for; the ranges below are therefore approximate. 300 samples were integrated over 1900–2150 and the eclipse switches recorded.

- **End of the eclipses:** the last switch before 2026 lies in 2000.0–2004.0–2006.5 (16–50–84 %). The data fix it to within a few years.
- **Return:** 2071.9–2110.0–2130.1, and 12 of the 300 samples (4 %) do not return before 2150. The return is not fixed by the data: χ^2 is flat in P_3 from about 270 to 400 d, and the nodal period, which sets the return, changes with it.
- **Third star:** m_3 2.5–3.9–5.2 $M_\odot$, outer inclination 61.9–66.6–72.5°.

The best fits (end 1999–2005.5, return 2058–2071) are therefore single points of a wide posterior. The end of the eclipses around 2000–2006 is a result; the return is not, anything from the early 2070s to after 2150 agrees with the data.

Table 8. QX Cas: the papers and SWINGERsim.

quantity	Guinan et al. (2012a,b)	SWINGERsim
inclination ~1960, ~1980	85.4°, 82°	85.6°, 81.8 ± 1.5°
eclipses absent	mid-1990s (Henden), 2003–2008 (FCAPT)	after 1999 geometrically; deeper than 0.02 mag only until 1994.5
fall of the inclination	~0.23°/yr (85.4° in ~1960 to ~76° from 2003)	~0.5°/yr in 1980, 0.7–0.8°/yr around 2000; nodal period 137 yr

quantity	Guinan et al. (2012a,b)	SWINGERsim
third star	unseen	ensemble: P_3 190–1030 d, m_3 0.5–3.9 $M_\odot$; fit: 288 ± 1 d, 4.6 ± 0.6 $M_\odot$
third light	not stated	needed for the old depths: 13–47 % (Table 6); a 4.6-$M_\odot$ third star gives ~12–14 %
eccentricity of the pair	0.21	0.23
photographic minima 1933–1942	observed (VarAstro)	geometric fit: not detectable before 1943; deep-1930s fit: detectable from 1936.5, but eclipses then last until 1999.0
eclipses return	–	geometrically 2058–2071 (the two fits); MCMC of the deep-1930s fit: 2072–2110–2130, 4 % not before 2150
times of minima: rms	–	photoelectric 45 min, photographic 98 min, visual 107 min

The inclination must fall faster than the published linear rate. In SWINGERsim the eclipse limit of the pair (from $R_1 + R_2 = 0.27$ a, the fractional radii of Guinan et al. 2012a, and the eccentric orbit) is about 70.5°. If the inclination was 82° in 1980 and there were no eclipses by 2003, it must have fallen below 70.5° within 23 years, i.e. at least 0.5°/yr on average. The fit does exactly that: the inclination peaks around 1965 and then falls ever faster. A steady 0.23°/yr would have kept QX Cas eclipsing until about 2030. The published value of about 76° from 2003 onwards comes from a light-curve model in which 76° gives no eclipses, whereas in our model a geometric eclipse would continue down to 70.5°. Our depth calculation shows how the two can be reconciled: at 76° the model eclipse is only about 0.06 mag deep, and FCAPT, which saw none, would have detected that, so 76° is an upper limit in our model as well. We therefore used it only as an upper limit (75 ± 1°). The ensemble above used the slower linear rate; a faster rate moves the third star closer or makes it heavier, so the ensemble range is, if anything, too wide on the long-period side.

Why the fitted third star is heavier than the ensemble median. The fitted period of the third star (288 d) is set mainly by the precession rate and the prior, because the 96 old minima scatter by one to two hours and cannot measure a light-time effect of a few minutes. A heavier third star at the same period turns the orbit faster, which the fit uses to make the inclination fall fast enough. The ensemble range (190–1030 d, 0.5–3.9 $M_\odot$) is the result; the fit is one concrete system that satisfies everything known about the inclinations.

4.6 A prediction tested: the binary seen without eclipses

An eccentric binary varies in light even without eclipses. Near periastron the two stars are closest: their tidal distortion (ellipsoidal variation) and the heating of one star by the other are largest, and the system brightens briefly once per orbit (a “heartbeat”; Thompson et al. 2012). SWINGERsim computes this from the same model that gives the orbits. For QX Cas today it predicts a variation of about 1 ppt, repeating every 6.0 days.

Why most of the signal is at half and a third of the period. A star distorted by tides is elongated towards its companion. We see it largest twice per orbit (when we look at the long side, at the quadratures) and smallest twice (at the conjunctions), so the ellipsoidal variation has two maxima per orbit: its period is P/2 (the second harmonic of the orbital frequency). In an eccentric orbit the distortion is strong only during the short passage through periastron and weak for the rest of the orbit. A brief brightening once per orbit is not a sine wave. Written as a sum of sine waves (a Fourier series) it needs the orbital period P together with its harmonics P/2, P/3, P/4 …, and for $e \approx 0.2$ the power is largest at P/2 and P/3. The periodogram therefore shows the binary most strongly at 3.0 and 2.0 days, not at 6.0 days.

We searched the TESS light curves of 2019–2024 (eight SPOC sectors between 18 and 85, about 200 days in all, 30-min bins) for this signal. QX Cas lies in a cluster, and light of

neighbouring stars leaks into the photometry: six periodic signals unrelated to the binary (1.04, 1.72, 3.18, 5.35, 5.63 and 6.43 d) were removed first. The same steps were applied to the SWINGERsim model light curve, so that data and model were treated identically.

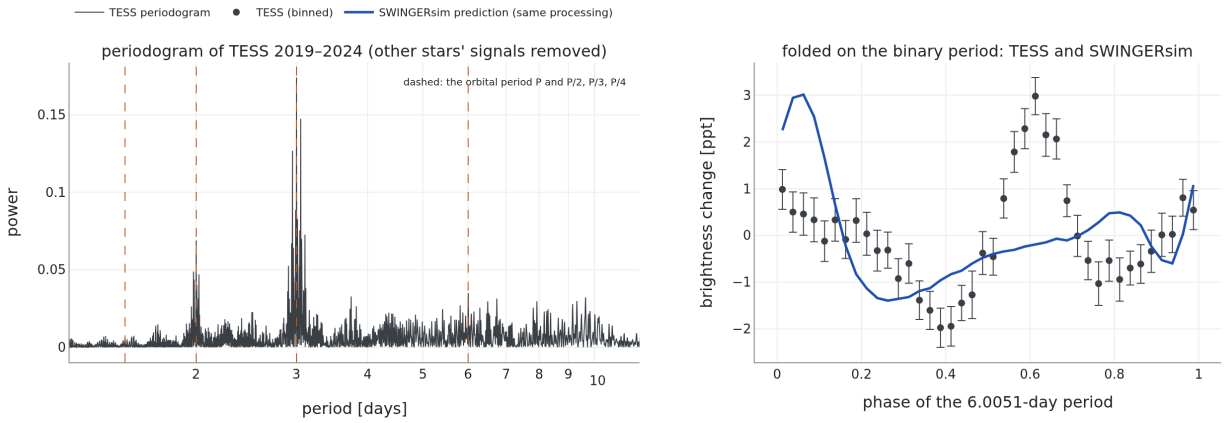


Figure 11. QX Cas in TESS 2019–2024. Left: periodogram after removing the signals of other stars; dashed: the binary period P = 6.0039 d and P/2, P/3, P/4. Right: TESS folded on the binary period (points, binned) and the SWINGERsim prediction processed in the same way (line).

How to read Figure 11. The left panel shows how strongly each period is present in the light. The highest peaks lie at P/2 and P/3 of the binary period: the binary is still visible in the light. The right panel folds all eight sectors on the binary period and averages them in 40 bins.

Table 9. QX Cas: amplitude of the binary signal in TESS and in the SWINGERsim prediction (same processing). Noise: 99 % of random periods give less than 0.47 ppt.

harmonic	period [d]	TESS [ppt]	SWINGERsim [ppt]
1	6.005	0.55 ± 0.14	0.98
2	3.003	**1.22 ± 0.09**	0.84
3	2.002	**0.81 ± 0.09**	0.72
4	1.501	0.17 ± 0.06	0.53

The binary is detected without eclipses, at the predicted level. The highest peak of the whole periodogram (1.2–12 d) lies at 3.0019 d, half of 6.0038 d, while the orbital period from the eclipses of the 20th century is 6.0039 d. The second and third harmonics are 14σ and 9σ signals, with amplitudes within about 30 % of the model's.

The shape is not yet right. The measured brightening is less sharply peaked than the model's and sits at a different phase (Fig. 11, right). The phase is not expected to agree: the model extrapolates the times of periastron from minima of 1933–1974 over about 2900 orbits, and the period error of the fit (±0.0002 d) alone shifts the phase by ±0.1 orbit; the orientation of the orbit (ω) after 50 years of apsidal motion is equally uncertain. The shape depends on ω, on the inclination and on the stellar parameters, which (see above) are not calibrated on the old eclipses. The TESS light curve therefore carries information on the present orbit that the old minima cannot give, and a fit of it is the natural next step.

4.7 Verdict

- **Confirms** that a third star can turn the orbit of QX Cas at the observed rate and switch off its eclipses in the 1990s. A stable system that does so exists and satisfies all inclination information, including the absence of eclipses in 2003–2024.

- **Adds:** where the third star can be: most likely $P_3 \approx 0.5$–3 yr, $m_3 \approx 0.5$–4 $M_\odot$, probably not on a nearly perpendicular orbit. Such a star moves the pair by a few to 25 km/s in radial velocity over months to years, which spectra can test.
- **Adds:** the binary is detected in TESS without eclipses, at its orbital period and with the predicted amplitude of the variation near periastron.
- **Differs:** the inclination must have fallen at least twice as fast as the published linear rate ($\geq 0.5°$/yr after 1980), and in our model the eclipses become undetectable in 1987.5–1994.5 rather than later.
- **Open problem, found by computing depths:** the photographic minima of 1933–1942 need an inclination of at least ~76–78° then. A fit that enforces this exists (75.8° in 1933, eclipses detectable on plates from 1936.5), but its eclipses then remain detectable until 1999.0, in tension with the absence of eclipses in the mid-1990s. The old eclipse depths also need third light that changes between eclipses and epochs. Both point to the adopted stellar radii and light ratio: they must be fitted together with the old depths and the TESS light curve. The DASCH plates can test the 1930s independently.
- **Dates:** an MCMC of the deep-1930s fit puts the end of the geometric eclipses in 2000.0–2006.5 (16–84 %). Their return is not constrained: 2072–2130, and 4 % of the samples not before 2150.
- **Caution:** the inclinations of ~1960 and ~1980 are read from Guinan et al. (2012a); their errors are our assumptions, and the underlying solutions (Bonaro et al. 2009) are unpublished. SWINGERsim therefore gives a constraint on the third star, not a detection.

5 AH Cep

5.1 What the papers say

AH Cep is one of the most massive eclipsing binaries in the Galaxy: two B0.5 V stars of 16.14 ± 0.26 and 13.69 ± 0.21 $M_\odot$ on a circular 1.775-d orbit, inclined at $69.81 \pm 0.60°$, with fractional radii 0.340 and 0.295, close to filling their Roche lobes (Pavlovski et al. 2018). That solution comes from the Strömgren uvby light curves of Bell et al. (1986), with the light ratio of the two stars fixed at the spectroscopic value 0.715 ± 0.020 and no third light; the spectra allow at most 2 % of third light for a companion with similar lines. The eclipse times since 1901 show a long wave, interpreted as the light-time effect of a third body on a ~68-yr orbit with $e \approx 0.53$ and a semi-amplitude of about 0.059 d; a second, ~9-yr light-time orbit of a possible fourth body was also proposed (Kim et al. 2003, 2005). Drechsel et al. (1989) reported a long-term change of the light-curve amplitude, which would mean a change of the inclination; Pavlovski et al. (2018) call this claim “uncorroborated”, i.e. not supported by later work, but did not test it themselves.

The question for SWINGERsim: is the wave in the eclipse times a third body, what can it be, can it or a fourth body change the inclination of the binary, and what do the TESS light curves say?

5.2 The system: published and fitted

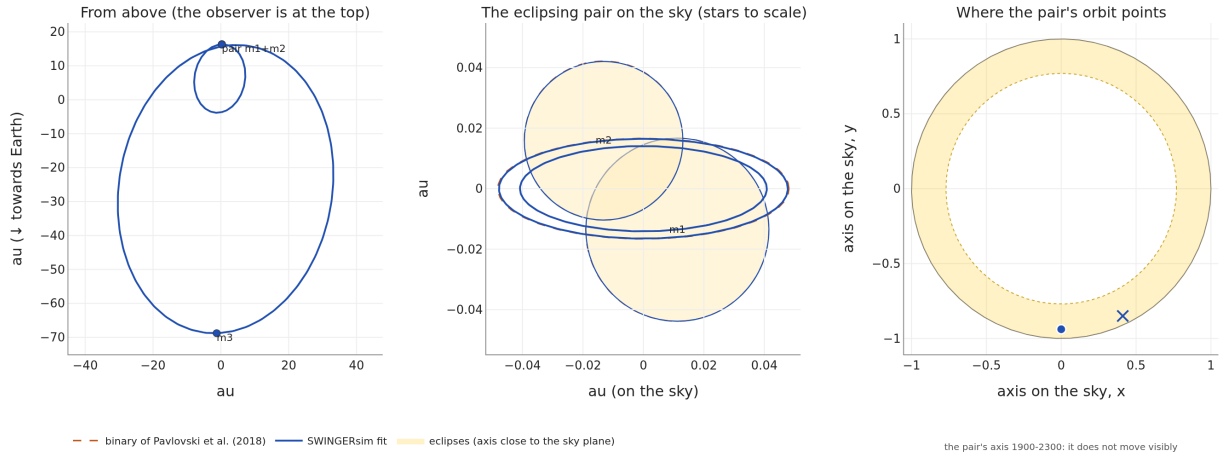


Figure 12. AH Cep: the binary of Pavlovski et al. (2018) (orange, dashed) and the SWINGERsim fit with the third body (blue). Left: from above – the pair and the third body on its 70-yr orbit; middle: the pair on the sky, stars to scale; right: the axis of the pair's orbit, which does not move visibly in 400 years.

How to read Figure 12. The scale of the left panel is tens of au: the pair moves on the small ellipse around the common centre of mass, the third body on the large, very eccentric one. In the middle panel the two stars nearly fill the space between them, a near-contact binary. In the right panel the axis of the pair sits in the eclipsing band and does not move: the third body is too far away to turn the pair's orbit within centuries.

5.3 125 years of eclipse times

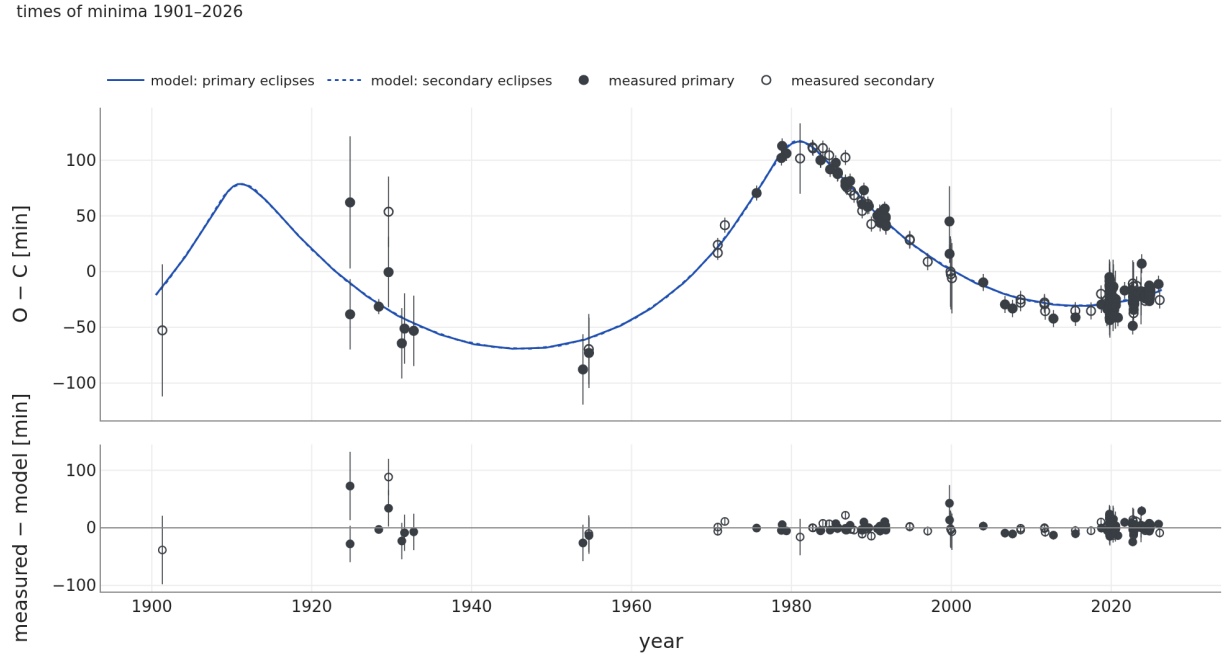


Figure 13. AH Cep: times of minima 1901–2026 against a fixed period (top) and minus the model (bottom). Line: model eclipse times; points: measured minima (visual, photographic, photoelectric, CCD, TESS).

How to read Figure 13. The eclipses come up to an hour and a half late or early compared with a strict clock. This is the light-travel time across the orbit of the pair around the third body: the eclipses are late when the pair is on the far side of that orbit (around 1980). The steep part in the 1970s is the passage of the eccentric outer orbit through periastron, when the pair moves fastest. Here the dynamical effect of the third body is negligible (the outer orbit is 14 000 times longer than the inner one). The model follows the TESS minima to 0.6–1 min in the best sectors (S18, S78) and the 125-year wave to the precision of each source.

Table 10. AH Cep: the papers and SWINGERsim.

quantity	papers	SWINGERsim fit
masses of the pair	16.14 ± 0.26, 13.69 ± 0.21 $M_\odot$ (Pavlovski et al. 2018; used as priors)	15.85 ± 0.19, 13.55 ± 0.19 $M_\odot$
outer period	~68 yr (Kim et al. 2003, 2005)	**69.7 ± 0.9 yr**
outer eccentricity	~0.53 (Kim et al. 2003)	**0.63 ± 0.03**
light-time semi-amplitude	~0.059 d (Kim et al. 2003)	0.058 d
third body	–	**$m_3 \sin i_3 \approx 6.6\ M_\odot$, so $m_3 \geq 6.6\ M_\odot$**
change of the inclination	claimed (Drechsel et al. 1989), called uncorroborated (Pavlovski et al. 2018)	**0.02° in 400 years** from the third body
stability	–	very hierarchical (Mardling–Aarseth ratio ≥ 52 for any mutual inclination)
times of minima: rms	–	TESS 0.6–2.6 min (best sectors), CCD 8.5 min, photoelectric 6.6 min, visual 31 min

The eclipse times measure the size of the pair's orbit around the third body and its period. From these follows the mass function, which for the masses of the pair gives $m_3 \sin i_3 \approx 6.6\ M_\odot$. Mass and orbital inclination of the third body are not separable (in the fit their correlation is −1.000): any $m_3 \geq 6.6\ M_\odot$ with the matching i_3 fits equally well.

5.4 A fourth body?

Can a fourth body, hidden in the eclipse times, tilt the pair? An ensemble of 1500 fourth bodies of 0.3–8 $M_\odot$ on orbits around the pair (period 100–5000 d, eccentricity 0–0.6, arbitrary orientation), with the third body as fitted, was integrated 20 years forward and backward. Only systems stable by Mardling & Aarseth (2001) were kept; with the 70-yr eccentric third body this allows periods up to about 2.8 years only. For each system the light-time effect of the pair on the fourth body's orbit was added to the residuals of the three-body fit; if χ^2 became worse by more than 25 (after refitting a quadratic ephemeris), the fourth body would have been seen.

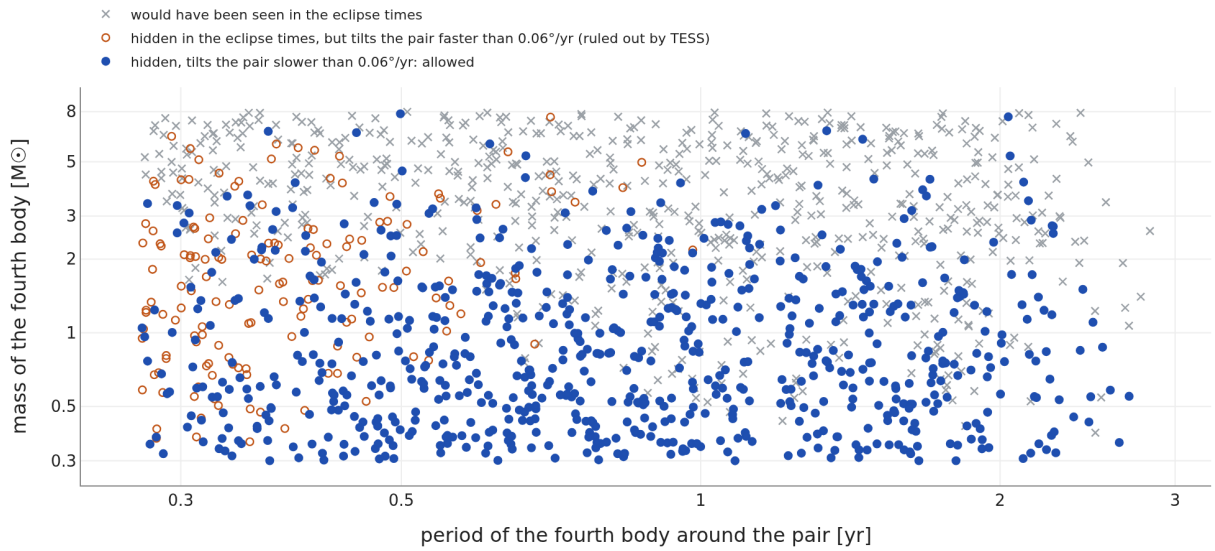


Figure 14. AH Cep: possible fourth bodies on stable orbits around the pair. Grey ×: their light-time effect would show in the eclipse times; orange circles: hidden in the eclipse times, but they would tilt the pair faster than the TESS eclipse depths allow; blue dots: hidden and slow enough.

How to read Figure 14. Each symbol is one possible fourth body. Massive ones (> 3 $M_\odot$) are mostly excluded by the eclipse times (84 %), light ones (< 1 $M_\odot$) mostly hidden (89 %); the largest hidden light-time signal is ±3 min. 859 of the 1500 bodies (57 %) are hidden in the eclipse times. A hidden fourth body can still tilt the pair: 153 of them (18 % of the hidden ones, mostly at periods below half a year) would change the inclination by more than 0.06° per year today. The constant TESS eclipse depths (next section) rule those out.

Two conclusions follow. First, the ~9-yr fourth body of Kim et al. (2003, 2005) cannot orbit the pair: with the 68–70-yr third body on an orbit of e ≈ 0.5–0.6, a 9-yr orbit around the pair is unstable by the criteria of Mardling & Aarseth (2001) and Vynatheya et al. (2022) (stability ratios 0.3–0.6 instead of > 1) for any mass and orientation. Second, a fourth body is not needed by the eclipse times, and if it exists it changes the inclination of the pair by less than about 0.06°/yr today.

5.5 What the TESS light curves say: the inclination and a third light

A main-sequence star of ≥ 6.6 $M_\odot$ of the age of the pair (about 6 Myr) is an early-B star. From MIST models (Choi et al. 2016) one star of 6.6–7 $M_\odot$ would contribute **7.0–7.9 %** of the TESS light, and a pair of 3.5 + 3.3 $M_\odot$ stars with the same total mass 3.2 %. Such light dilutes the eclipses.

We asked what the TESS eclipses say in two ways.

1. **The published pair, inclination scanned.** For inclinations between 69.8° and 76° we computed the light curve of the published pair (radii and temperatures of Pavlovski et al. 2018, Roche surfaces) and fitted sectors 77 and 78 (2024) and 18 (2019) with a free constant third light.
2. **A fit with the stars free.** A SWINGERsim fit of sectors 77 and 78 with the inclination, both radii and the temperature of the secondary free, Roche surfaces, the 216 TESS minima for the timing, the correlated noise of the light curve taken into account (red-noise factor β ≈ 2.2; Pont et al. 2006), and the depth of every light-curve segment scaled by a free factor. The model contains the light of a 7 $M_\odot$ third star, so the scale factor measures how much third light the eclipses need.

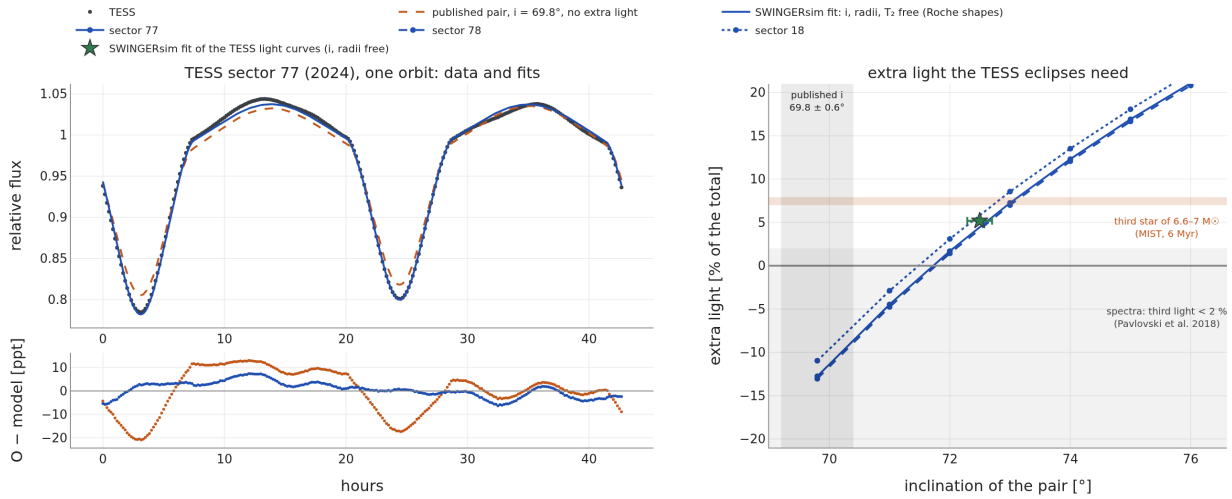


Figure 15. AH Cep in TESS. Left: one orbit of sector 77 (2024) with the published pair at i = 69.8° without third light (orange, dashed) and the SWINGERsim fit with the stars free (blue); bottom: residuals. Right: the third light the TESS eclipses need for the published pair at each assumed inclination (lines; three sectors), the SWINGERsim fit (star), the light of a 6.6–7 $M_\odot$ third star (orange band) and the range allowed by the spectra (grey).

How to read Figure 15. Left: the published pair at 69.8° gives eclipses that are too shallow and too broad; the fit (blue) matches the light curve to 3.0–7.1 ppt rms. The small changes of slope where an eclipse begins and ends (around 7 and 20 hours) are the contacts of the two stars on the sky: there the eclipsed area starts or stops growing. The remaining wiggles of a few ppt are partly instrumental (they repeat at the ends of the TESS segments). Right: each line says how much light must be added (positive) or removed (negative) for the published pair to match the TESS eclipse depths at a given inclination. At 69.8° light would have to be removed, which is impossible; without any third light the depths need about 71.7°, and every percent of third light raises this by about 0.2°. The fit with free stars gives **i = 72.50 ± 0.21°** (formal), R_1 = 6.25 ± 0.05 $R_\odot$ and R_2 = 5.88 ± 0.05 $R_\odot$ (published 6.51 and 5.64 $R_\odot$), T_2 = 28 430 ± 120 K, and **about 5 % of third light**.

Table 11. AH Cep: the eclipses of the 1980s and of TESS.

	Pavlovski et al. (2018)	SWINGERsim, TESS 2024
data	uvby, Bell et al. (1986)	TESS sectors 77, 78
inclination	69.81 ± 0.60°	72.50 ± 0.21° (formal)
radii	6.51, 5.64 $R_\odot$	6.25 ± 0.05 $R_\odot$, 5.88 ± 0.05 $R_\odot$
light ratio of the stars	0.715 ± 0.020 (spectra, fixed)	0.78 in B (published radii give 0.67 in the same model)
third light	0 (spectra: < 2 %)	about 5 % (fitted)
masses ($\propto 1/\sin^3$ i)	16.14, 13.69 $M_\odot$	15.4, 13.0 $M_\odot$ if the same velocities are combined with this inclination

The TESS eclipses did not change in five years. At a fixed inclination, sector 18 (2019) needs about 1.3 % more third light than sectors 77 and 78 (2024). Since the needed third light changes by about 4.8 % per degree of inclination, this corresponds to at most 0.27° in 4.4 years, i.e. a rate $|di/dt| \leq 0.06°$/yr; part of the difference is the sector-to-sector calibration of TESS for this bright star.

Is the inclination of AH Cep different from the published one? Three explanations remain open:

- the inclination is higher and there is a few percent of third light, the light of the ≥ 6.6 $M_\odot$ third body. The masses of the pair would then be lower. The spectra limit the third light to < 2 % for a star with lines similar to those of the pair, which a 21 000 K B2 star would have; very broad lines of a fast rotator could make it harder to see;
- the third body is fainter for its mass (two lighter stars, a compact object), and the radii of the pair differ from the published ones;
- the inclination did not change, but the light-curve models differ. For the same parameters SWINGERsim's eclipses are about 5 % shallower than PHOEBE's in V, and for V907 Sco the same tendency shifts SWINGERsim's inclination upwards by a fraction of a degree. For AH Cep the depth of each segment was left free, so the inclination rests on the shape of the light curve (the widths of the eclipses and the ellipsoidal variation), which depends on the Roche model, gravity darkening and limb darkening.

A change of the inclination between the 1980s and 2024 cannot be caused by the third body (0.02° in 400 years), and a hidden fourth body could give at most about 0.06°/yr today. An independent analysis of the TESS light curves with PHOEBE or the Wilson–Devinney code, with free third light, is the decisive test.

5.6 Verdict

- **Confirms** the light-time orbit of Kim et al. (2003, 2005): one third body on a 70-yr eccentric orbit reproduces 125 years of eclipse times, from visual estimates to TESS, with the same semi-amplitude.
- **Disagrees** with a dynamical change of the inclination (Drechsel et al. 1989): the third body changes the inclination by 0.02° in 400 years. This supports the view of Pavlovski et al. (2018), who call the claim uncorroborated, with a dynamical argument that they did not make.
- **Adds:** the third body must be massive (≥ 6.6 $M_\odot$); as a single main-sequence star it would give 7.0–7.9 % of the light. A fourth body is not needed; the ~9-yr fourth body proposed earlier would be dynamically unstable, and a hidden fourth body could tilt the pair by at most ~0.06°/yr today.
- **Adds, in tension with the spectra:** the TESS eclipses (2019, 2024) are best described by i ≈ 72.5° and about 5 % third light, in agreement with the predicted third star but not with the published inclination and the spectroscopic limit on third light. This result depends on the light-curve model and needs an independent check.
- **Can be checked with Gaia and Hipparcos:** the 70-yr orbit accelerates the pair on the sky (see below).

6 Independent checks with Gaia

Gaia measures the position of the light centre of a star many times over three years. If the light centre moves on an orbit, the single-star solution fits worse: the renormalised unit weight error (RUWE) grows above about 1.4 (the threshold proposed by Lindegren 2018), the astrometric excess noise becomes significant, the parallaxes of DR2 and DR3 (different time spans) may disagree, and for clear cases the Non-Single Star catalogue gives an astrometric orbit. These are independent of everything used above. A third star of mass m_3 and light fraction l_3 on an orbit of size a around a pair moves the light centre by about a ($m_3/M - l_3$), where M is the total mass.

Table 12. The expected motion of the light centre from the models of this report. The distances are rough values from the brightness of the stars; Gaia's parallax will replace them.

	V907 Sco	QX Cas	AH Cep
outer orbit	142 d, a ≈ 0.98 au	190–1030 d, a ≈ 1.6–4.8 au (fit: 288 d, 2.2 au)	70 yr, a ≈ 56 au
$m_3/M - l_3$	0.16 − 0.02 = 0.14	0.04–0.17 (fit: 0.28 − 0.13 = 0.15)	≥ 0.19 − 0.07 ≈ 0.12

	V907 Sco	QX Cas	AH Cep
light centre moves by	≈ 0.14 au	≈ 0.06–0.8 au (fit: 0.3 au)	≈ 7 au over 70 yr
at a distance of	0.3–0.5 kpc: ≈ 0.3–0.5 mas	~3.3 kpc (NGC 7790): 0.02–0.25 mas (fit: 0.1 mas)	~1 kpc: ≈ 7 mas; the proper motion changes by ~1 mas/yr in 25 yr
expected in Gaia	RUWE > 1.4 likely; NSS orbit possible	at the limit	a proper-motion difference Hipparcos–Gaia

V907 Sco should show the third star clearly; QX Cas is at the limit of what Gaia can see, so a normal RUWE would not exclude the ensemble; AH Cep, bright enough for Hipparcos, should show the 70-yr orbit as a change of its proper motion between Hipparcos (1991) and Gaia (2016).

6.1 What Gaia DR3 shows

We queried the Gaia archive (DR3, Gaia Collaboration et al. 2023, and for comparison DR2) at the positions of the three systems with `tools/gaia_check.py`. The astrometry of DR3 is the solution described by Lindegren et al. (2021).

Table 13. Gaia DR3 astrometry of the three systems. Columns:

- *G*: Gaia magnitude;
- *parallax*: DR3 parallax with its error, and the corresponding distance (1/parallax);
- *DR2 parallax*: the value in DR2 and the difference DR3 − DR2 in units of the combined error;
- *RUWE*: values above about 1.4 indicate a poor single-star fit;
- *excess noise*: the astrometric excess noise (AEN, the extra scatter needed to fit the positions) and its significance;
- *NSS*: the solution type in the Non-Single Star catalogue, if any.

system	G	parallax [mas] (distance)	DR2 parallax (difference)	RUWE	excess noise (signif.)	NSS
V907 Sco	8.55	2.279 ± 0.033 (439 pc)	2.337 ± 0.064 (−0.8σ)	0.81	0.16 mas (11)	SB2, P = 3.978 d
QX Cas	10.12	0.268 ± 0.019 (3.7 kpc)	0.115 ± 0.031 (+4.2σ)	1.18	0.17 mas (39)	–
AH Cep	6.84	1.139 ± 0.024 (878 pc)	1.102 ± 0.034 (+0.9σ)	1.18	0.23 mas (52)	eclipsing binary, P = 1.774737 d

How to read Table 13. Each row is one system; the columns are explained above. None of the three has a RUWE above 1.4. The excess noise is formally significant in all three, but for bright stars this is the rule rather than the exception: among the five-parameter solutions of Gaia EDR3 (the same astrometry as DR3) with G = 9–12, about 97 % have a significant excess noise, with a median of about 0.11 mas (Gaia EDR3 online documentation, Sect. 4.5.2). Much of it is likely to come from imperfect calibration of bright stars rather than from orbital motion. A significant excess noise of 0.16–0.23 mas, as here, is therefore weak evidence on its own; the RUWE is the more robust indicator.

- **V907 Sco: a tension.** At the Gaia distance of 439 pc our model moves the light centre by about 0.14 au, i.e. about 0.32 mas, on the 142-d orbit. Gaia covered about seven such orbits within the DR3 time span. A motion of 0.3 mas at G = 8.5 usually raises the RUWE above 1.4, yet it is 0.81. The excess noise (0.16 mas) is of the right order, but it

does not by itself show an orbit. The NSS catalogue gives a double-lined spectroscopic (SB2) solution with P = 3.978 d. This is 5 % longer than the eclipse period of 3.776 d, so it probably reflects the inner pair seen with a poorly determined period, not the 142-d orbit. Possible reasons for the low RUWE are a smaller light-centre motion than modelled (more light from the third star, or a smaller m_3/M), or a cancellation by the scanning geometry. The third star itself is established from the eclipse times and radial velocities (Zasche et al. 2023), so this does not question it. But an astrometric orbit in a future Gaia release (DR4, with epoch astrometry) is a direct test of our m_3.

- **QX Cas: consistent, not decisive.** The RUWE is normal. The excess noise of 0.17 mas is about the size of the light-centre motion of our fit (0.1 mas at 3.3 kpc), but for a G = 10 star in a cluster that is not evidence. The parallax changed by 4.2σ between DR2 and DR3. The DR2 value (0.115 mas, i.e. 8.7 kpc) is incompatible with the cluster NGC 7790 at about 3.3 kpc, while DR3 agrees with it. Orbital motion of the light centre can bias a parallax measured over a shorter time span, so this is a hint, but DR2 systematics are an equally likely cause.
- **AH Cep: as expected.** The RUWE is normal, as expected for an outer orbit of 70 years that Gaia sees as nearly linear motion. The NSS catalogue recognises the eclipsing binary with exactly the period of the inner pair (1.774737 d). The test proposed above, a change of the proper motion between Hipparcos and Gaia, uses the Hipparcos–Gaia catalogue of accelerations (Brandt 2021). We have not yet made it; it is listed in *Open questions and future work*.

7 The three systems compared

Table 14. Summary. One column per system, one row per question; *verdict* compares SWINGERsim with the papers.

	V907 Sco	QX Cas	AH Cep
eclipses come and go	yes: stop ~1920, return ~1955, stop ~1987, return 2019	stopped in the 1990s	always eclipsing
nodal period	65.5 yr (published 63.5 yr)	137 yr (fit)	millions of years
third body	0.98 $M_\odot$, 142.09 d	0.5–4 $M_\odot$, 190–1030 d (ensemble)	≥ 6.6 $M_\odot$, 69.7 yr
verdict	**confirms** the published solution; adds: the 99-d orbit ruled out, the orbit change seen in TESS minima	**confirms** the interpretation; adds: where the third star is, the binary detected in TESS without eclipses; the inclination fell faster; open: the 1930s minima and the old depths	**confirms** the third body; **disagrees** with a changing inclination; adds: a fourth body not needed (the 9-yr one unstable), TESS needs a higher inclination and third light
prediction	end of the eclipses 2052–2057	geometric eclipses return ~2058–2071 (two fits)	constant eclipses
Gaia DR3	RUWE 0.81: less light-centre motion than modelled (tension)	RUWE normal; DR2–DR3 parallax differs by 4.2σ	RUWE normal; Hipparcos–Gaia test open
what it rests on	14 inclinations, 172 velocities, 58 TESS minima	2 inclinations, limits, 96 old minima	325 minima, TESS light curves

8 Discussion

8.1 What makes these results reliable, and where they are weaker

- **V907 Sco** rests on the best data: inclinations from four independent sources, velocities from two epochs 40 years apart, precise TESS minima, and a posterior sampled with MCMC. The SWINGERsim result agrees with the published one within the errors, and SWINGERsim reproduces the published predictions when given the published parameters. The orientation of the orbits on the sky (the nodes) is not given in the paper; only the difference of the nodes enters any observable.
- **QX Cas** rests on two inclinations, one upper limit and the absence of eclipses. The ensemble is therefore the main result, and the fit is an example. Computing the eclipse depths showed that the fit satisfies the 1930s only geometrically, not observably, and that the old depths need third light and different stellar parameters. The TESS detection of the binary is an independent check of the model's amplitude, but not yet of its orbit.
- **AH Cep** rests on the light-time effect alone for the third body. The third-light result depends on the light-curve model.

8.2 A lesson: "the model eclipses" is not "the eclipses were seen"

For all three systems the switching dates were first compared as geometric crossings of the eclipse limit. For V907 Sco this was enough, because its eclipses deepen quickly after the crossing and the model depths at the observed switch dates were checked (Table 2). For QX Cas the same comparison hid a real problem: a geometric eclipse in 1933 that is 0.0006 mag deep is no eclipse for a photographic plate. Whenever an observation says "eclipses were seen", the model has to give a depth above the detection limit of that observation, with the third light stated. We now enter such observations in that form.

8.3 The light-curve model

In all three systems SWINGERsim's synthetic light curves were compared with TESS. They reproduce the eclipses of V907 Sco to within 5 % in depth without being fitted, and the heartbeat amplitude of QX Cas. Where depths are fitted, SWINGERsim's eclipses for given parameters are systematically about 5 % shallower than PHOEBE's (V907 Sco, AH Cep). This shifts fitted inclinations upwards by a fraction of a degree for V907 Sco and probably by more for the shallow, partial eclipses of AH Cep. Results that rest on eclipse depths (the light-curve fit of V907 Sco, the inclination and third light of AH Cep, the depths and detectability of the QX Cas eclipses, the shape of the QX Cas signal) should be read with this in mind. The dynamical results (eclipse times, velocities, geometric switching dates) do not depend on it.

8.4 Errors

The errors are linear errors from the curvature of χ^2, except for V907 Sco, where the MCMC posterior errors are given; for V907 Sco both agree within 50 %. The MCMC chain covers about 19 autocorrelation times in total and about 9 after the burn-in half is discarded (see Data and methods); a chain of 400 steps gave errors within a median of 10 % of these, so we consider them correct to a few tens of percent. For strongly correlated parameters (for example mass and orbital inclination of the third body of AH Cep) linear errors understate the true range.

9 Conclusions

1. **V907 Sco:** SWINGERsim reproduces the switching eclipses from the dynamics alone. The third star (0.98 (+0.05/−0.04) $M_\odot$, 142.09 ± 0.04 d), the mutual inclination (25.8°) and the nodal period (65.5 yr) agree with Zasche et al. (2023). The model switches the eclipses off and on when they were observed to do so, predicts the 2021–2023 eclipse depths to within 5 %, and predicts the end of the present eclipses in 2052–2057.

2. **V907 Sco:** the radial velocities of 1983–1994 fit the 142-d orbit; the 99-d outer period is ruled out. The eclipse times are dominated by the dynamical effect of the third star.
3. **QX Cas:** a third star of 0.5–4 $M_\odot$ at 0.5–3 years turns the binary at the observed rate and switched off its eclipses in the 1990s; the inclination must have fallen at ≥ 0.5°/yr after 1980.
4. **QX Cas is detected in TESS without eclipses**, at the orbital period and its harmonics, with the amplitude SWINGERsim predicts.
5. **QX Cas, open:** the photographic minima of 1933–1942 need an inclination of ≥ 76–78° then. A fit that enforces this exists, but then the eclipses stay detectable until 1999.0, against the absence of eclipses in the mid-1990s; the old eclipse depths also need third light that a single value cannot supply. Both point to the adopted stellar radii and light ratio. An MCMC of that fit, conditioned on the whole inclination history, puts the end of the geometric eclipses in 2000.0–2006.5 (16–84 %) but leaves their return open (2072–2130, 4 % of the samples not before 2150). A joint fit of the depths and the TESS light curve and the DASCH plates are the next steps.
6. **AH Cep:** 125 years of eclipse times give a third body of ≥ 6.6 $M_\odot$ on a 70-yr orbit. It cannot change the inclination measurably; a fourth body is not needed, and the proposed 9-yr fourth body would be unstable. TESS suggests i ≈ 72.5° with about 5 % third light, which is in tension with the published inclination and the spectra and requires an independent light-curve analysis.
7. **Gaia DR3:** none of the three has a RUWE above 1.4. For V907 Sco this is a tension: the modelled light-centre motion of about 0.3 mas on the 142-d orbit should usually be visible; Gaia DR4 epoch astrometry can test it directly. For QX Cas the parallax changed by 4.2σ between DR2 and DR3, a hint but not evidence of orbital motion; for AH Cep the Hipparcos–Gaia acceleration remains to be checked.

10 Data and reproducibility

The configurations are in `configs/squadra_test3/`, the observed data in `data/observed/<system>/` (minima with their sources, inclinations and limits, radial velocities, TESS light curves), and the outputs of the fits and ensembles in `output/squadra_test3/<system>/`. TESS minima were measured with `tools/tess_minima.py`, the QX Cas depths with `tools/qx_cas_extra.py`, the Gaia data are queried by `tools/gaia_check.py`. The figures and all numbers quoted are produced by `python tools/squadra3_report.py` and stored in `docs/figures/squadra_test3/numbers.json`.

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